Basic of Data Science

Pitchaya Wiratchotisatian

Data Visualization

Visualization:

The activity of forming a mental model of something

- R. Spence, Information Visualization

Data Visualization

- What data looks like
- Expressing numerical relations
- Understanding data
- Compressing data
- Understandable
- Patterns
- Communications

Statistical data presentation

- Data are usually collected in a raw format and thus the inherent information is difficult to understand
- Therefore, raw data need to be summarized, processed, and analyzed
- However, no matter how well manipulated, the information derived from the raw data should be presented in an effective format, otherwise, it would be a great loss for both authors and readers



Statistical Round
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Statistical data presentation

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Data are usually collected in a raw format and thus the inherent information is difficult to understand. Therefore, raw data need to be summarized, processed, and analyzed. However, no matter how well manipulated, the information derived from the raw data should be presented in an effective format, otherwise, it would be a great loss for both authors and readers. In this article, the techniques of data and information presentation in textual, tabular, and graphical forms are introduced. Text is the principal method for explaining findings, outlining trends, and providing contextual information. A table is best suited for representing individual information and represents both quantitative and qualitative information. A graph is a very effective visual tool as it displays data at a glance, facilitates comparison, and can reveal trends and relationships within the data such as changes over time, frequency distribution, and correlation or relative share of a whole. Text, tables, and graphs for data and information presentation are very powerful communication tools. They can make an article easy to understand, attract and sustain the interest of readers, and efficiently present large amounts of complex information. Moreover, as journal editors and reviewers glance at these presentations before reading the whole article, their importance cannot be ignored.

Key Words: Data presentation, Data visualization, Graph, Statistics, Table.

Statistical data presentation

Techniques of data and information presentation in

- Text

  If one wishes to compare or introduce two values at a certain time point, it is appropriate to use text or the written language

- Tabular forms

  Table is the most appropriate when all information requires equal attention, and it allows readers to selectively look at information of their own interest

- Graphical forms

  Graphs allow readers to understand the overall trend in data, and intuitively understand the comparison results between two groups

Statistical data presentation

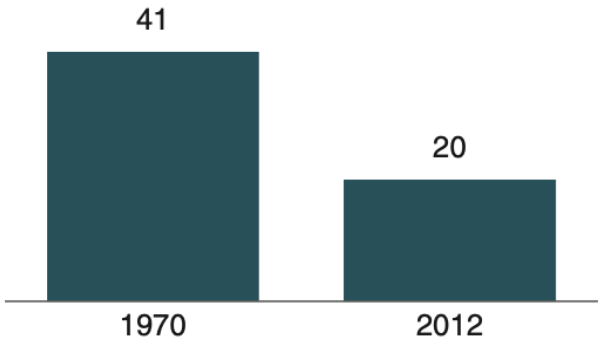
Text is the main method of conveying information as it is used to explain results and trends, and provide contextual information

- Information about the incidence rates of delirium following anesthesia in 2016–2017 can be presented with the use of a few numbers:
- “The incidence rate of delirium following anesthesia was 11% in 2016 and 15% in 2017; no significant difference of incidence rates was found between the two years.”

Text

Children with a "Traditional" Stay-at-Home Mother

% of children with a married stay-at-home mother with a working husband



Note: Based on children younger than 18. Their mothers are categorized based on employment status in 1970 and 2012.

Source: Pew Research Center analysis of March Current Population Surveys Integrated Public Use Microdata Series (IPUMS-CPS), 1971 and 2013

Adapted from PEW RESEARCH CENTER

FIGURE 2.2 Stay-at-home moms original graph

20%

of children had a
traditional stay-at-home mom
in 2012, compared to 41% in 1970

FIGURE 2.3 Stay-at-home moms simple text makeover

Statistical data presentation

Tables convey information that has been converted into words or numbers in rows and columns

- Tables have been used for nearly 2,000 years
- The strength of tables is that they can accurately present information that cannot be presented with a graph, i.e. “132.145852”

Table 1. Modified Table in Lee and Kim's Research (Adapted from Korean J Anesthesiol 2017; 70: 39-45)

Variable	Group	Baseline	After drug	1 min	3 min	5 min
SBP	C	135.1 ± 13.4	139.2 ± 17.1	186.0 ± 26.6*	160.1 ± 23.2*	140.7 ± 18.3
	D	135.4 ± 23.8	131.9 ± 13.5	165.2 ± 16.2*,†	127.9 ± 17.5†	108.4 ± 12.6†,‡
DBP	C	79.7 ± 9.8	79.4 ± 15.8	104.8 ± 14.9*	87.9 ± 15.5*	78.9 ± 11.6
	D	76.7 ± 8.3	78.4 ± 6.3	97.0 ± 14.5*	74.1 ± 8.3†	66.5 ± 7.2†,‡
MBP	C	100.3 ± 11.9	103.5 ± 16.8	137.2 ± 18.3*	116.9 ± 16.2*	103.9 ± 13.3
	D	97.7 ± 14.9	98.1 ± 8.7	123.4 ± 13.8*,†	95.4 ± 11.7†	83.4 ± 8.4†,‡

Values are expressed as mean ± SD. Group C: normal saline, Group D: dexmedetomidine. SBP: systolic blood pressure, DBP: diastolic blood pressure, MBP: mean blood pressure, HR: heart rate. *P < 0.05 indicates a significant increase in each group, compared with the baseline values. †P < 0.05 indicates a significant decrease noted in Group D, compared with the baseline values. ‡P < 0.05 indicates a significant difference between the groups.

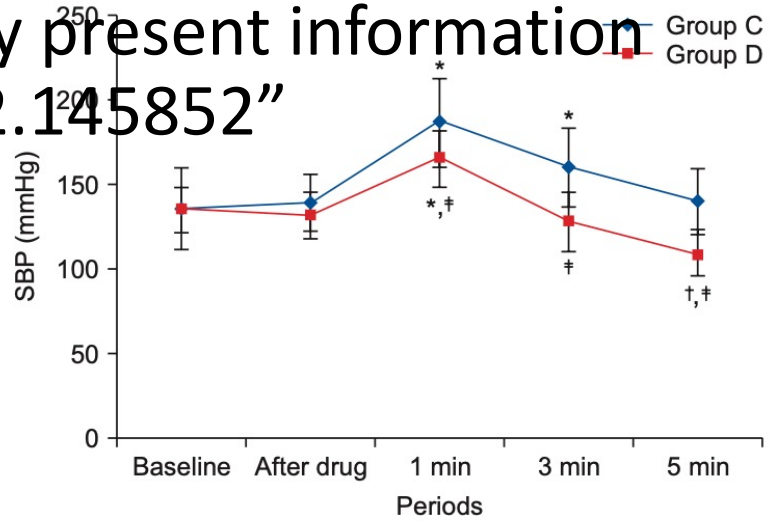


Fig. 1. Line graph with whiskers. Changes in systolic blood pressure (SBP) in the two groups. Group C: normal saline, Group D: dexmedetomidine. *P < 0.05 indicates a significant increase in each group, compared with the baseline values. †P < 0.05 indicates a significant decrease noted in Group D, compared with the baseline values. ‡P < 0.05 indicates a significant difference between the groups (Adapted from Korean J Anesthesiol 2017; 70: 39-45).

Table

- Data stands out more than the structural components of the table in the second and third iteration (light borders, minimal borders)

Heavy borders

Group	Metric A	Metric B	Metric C
Group 1	$X.X	Y%	Z,ZZZ
Group 2	$X.X	Y%	Z,ZZZ
Group 3	$X.X	Y%	Z,ZZZ
Group 4	$X.X	Y%	Z,ZZZ
Group 5	$X.X	Y%	Z,ZZZ

Light borders

Group	Metric A	Metric B	Metric C
Group 1	$X.X	Y%	Z,ZZZ
Group 2	$X.X	Y%	Z,ZZZ
Group 3	$X.X	Y%	Z,ZZZ
Group 4	$X.X	Y%	Z,ZZZ
Group 5	$X.X	Y%	Z,ZZZ

Minimal borders

Group	Metric A	Metric B	Metric C
Group 1	$X.X	Y%	Z,ZZZ
Group 2	$X.X	Y%	Z,ZZZ
Group 3	$X.X	Y%	Z,ZZZ
Group 4	$X.X	Y%	Z,ZZZ
Group 5	$X.X	Y%	Z,ZZZ

FIGURE 2.4 Table borders

Statistical data presentation

Tables convey information that has been converted into words or numbers in rows and columns

- Tables have been used for nearly 2,000 years
- The strength of tables is that the …on that cannot be presented with a

Table 2. Difference between a Regular Table and a Heat Map

Example of a regular table				Example of a heat map			
SBP	DBP	MBP	HR	SBP	DBP	MBP	HR
128	66	87	87	128	66	87	87
125	43	70	85	125	43	70	85
114	52	68	103	114	52	68	103
111	44	66	79	111	44	66	79
139	61	81	90	139	61	81	90
103	44	61	96	103	44	61	96
94	47	61	83	94	47	61	83

All numbers were created by the author. SBP: systolic blood pressure, DBP: diastolic blood pressure, MBP: mean blood pressure, HR: heart rate.

Statistical data presentation

Graphs simplify complex information by using images and emphasizing data patterns or trends

- Graphs are useful for summarizing, explaining, or exploring quantitative data
- A graph format that best presents information must be chosen so that readers and reviewers can easily understand the information

Scatter Plot

Scatter plots present data on the x- and y-axes and are used to investigate an association between two variables

- A regression line is added to a graph to determine whether the association between two variables can be explained or not

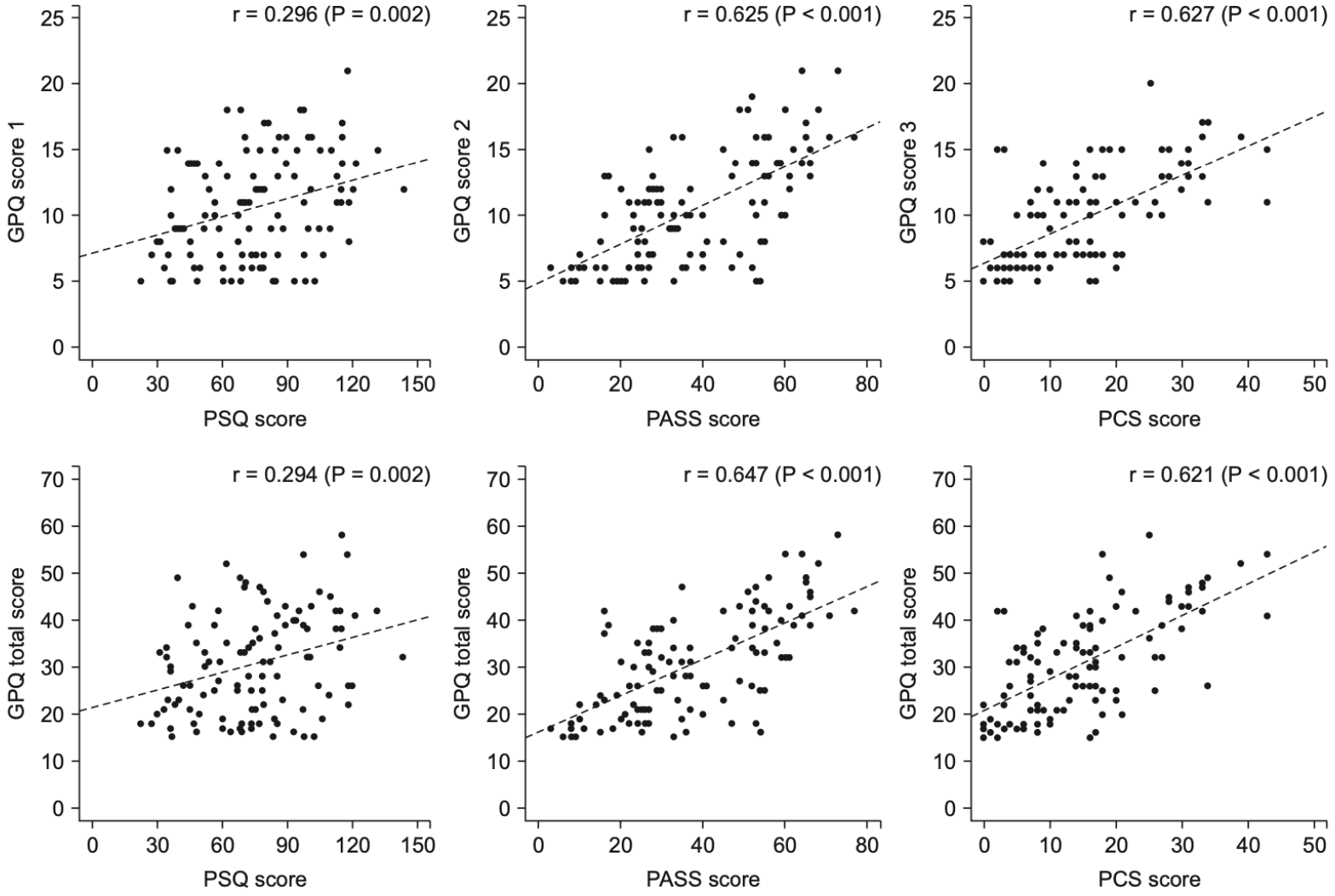


Fig. 2. Scatter plot of GPQ scores and the other questionnaires. Score 1: sensitivity, 2: experience, 3: other. GPQ: Geop-Pain Questionnaire, PSQ: Pain Sensitivity Questionnaire, PASS: Pain Anxiety Symptoms Scale, PCS: Pain Catastrophizing Scale (Adapted from Korean J Anesthesiol 2016; 69: 492-505).

Bar Graph and Histogram

- A bar graph compare values in a discrete category or group, and the frequency or other measurement parameters (i.e. mean)
- Bar graphs are flexible, and can be used in a grouped or subdivided bar format in cases of two or more data sets in each category
- It is advised to start the x- and y- axes from 0, illustration of comparison results in the x- and y- axes that do not start from 0 can deceive readers' eyes and lead to overrepresentation of the results

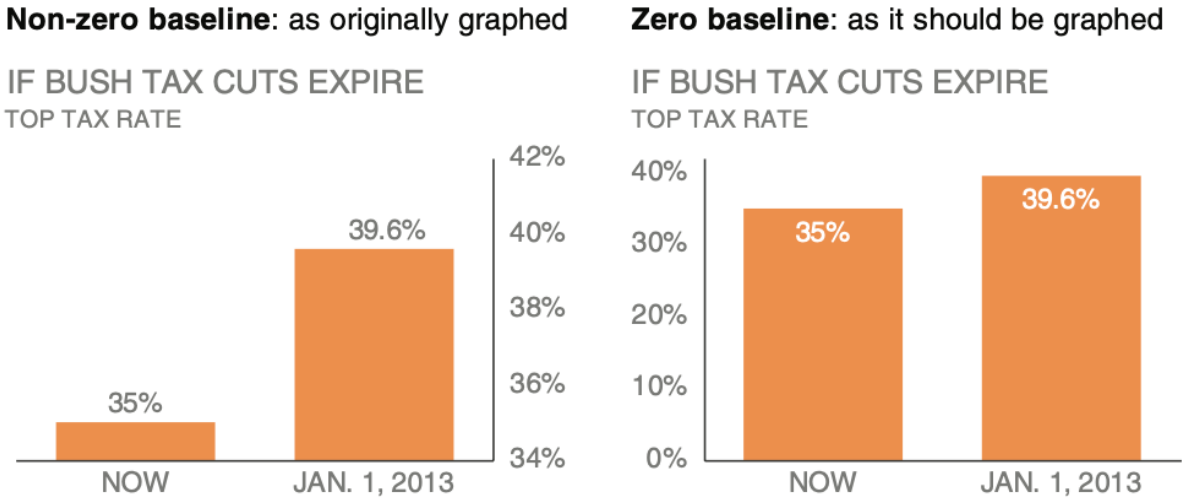


FIGURE 2.13 Bar charts must have a zero baseline

Bar Graph and Histogram



FIGURE 2.14 Bar width

Bar Graph and Histogram

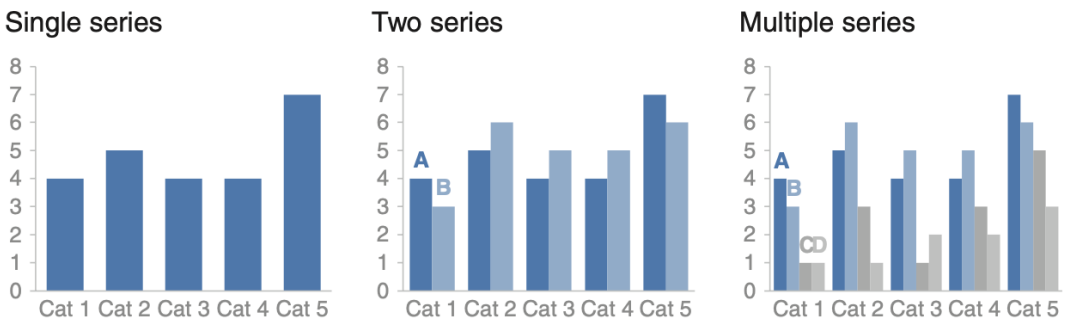


FIGURE 2.15 Bar charts

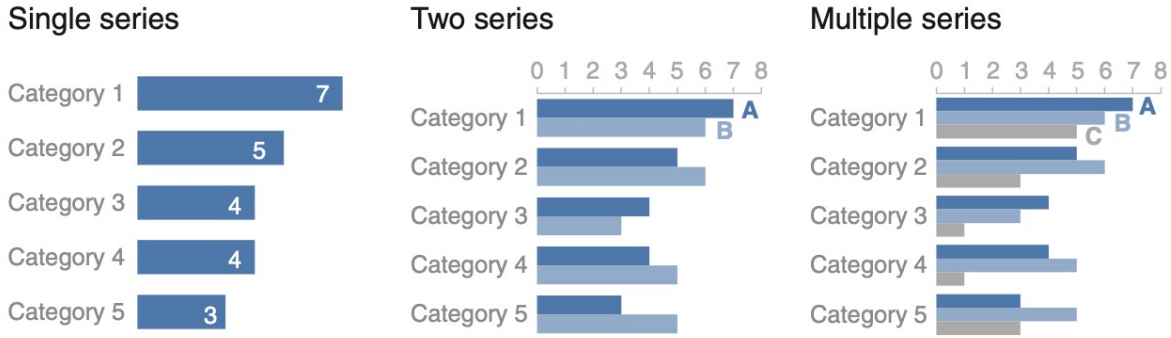


FIGURE 2.18 Horizontal bar charts

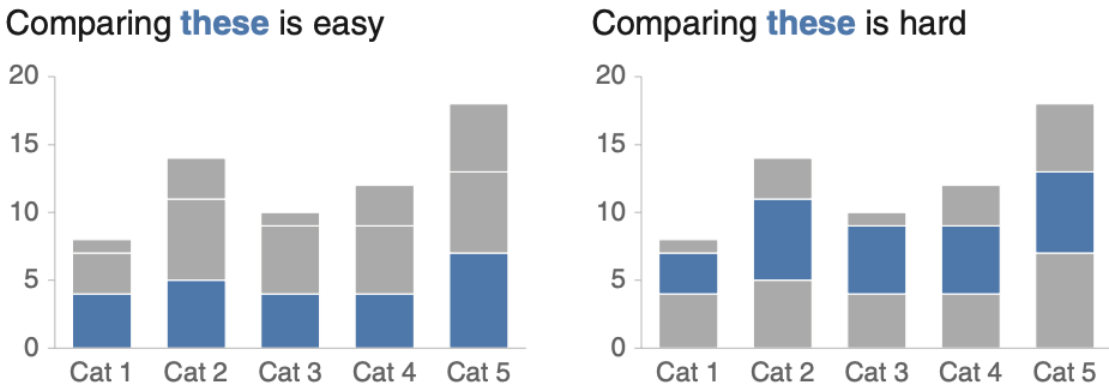


FIGURE 2.16 Comparing series with stacked bar charts

Bar Graph and Histogram

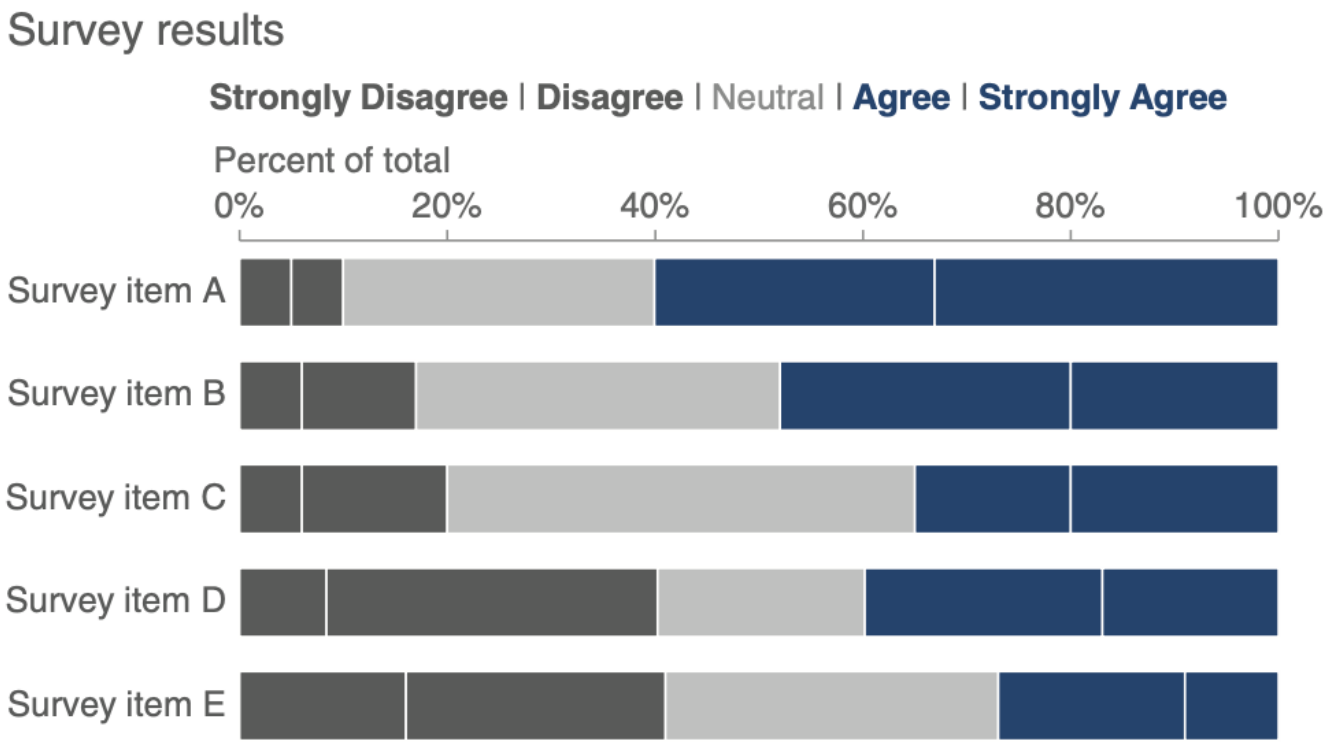


FIGURE 2.19 100% stacked horizontal bar chart

Bar Graph and Histogram

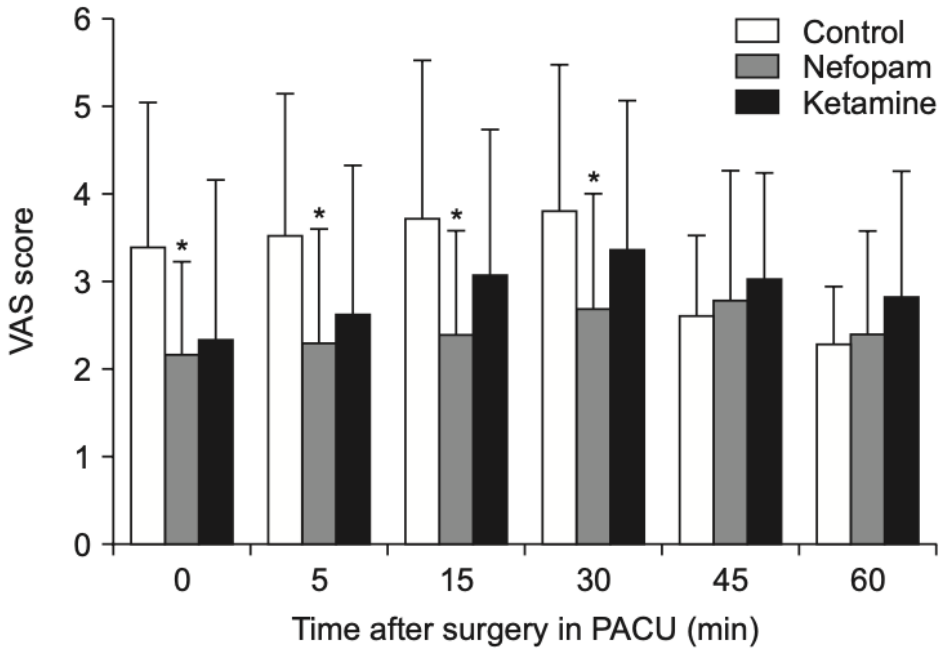


Fig. 3. Multiple bar graph with whiskers. Pain scores in the recovery room. *P < 0.05 compared with the control group. The nefopam group showed significant lower visual analogue scale (VAS) score at 0, 5, 15, 30, 45 and 60 minutes on postanesthesia care unit compared with the control group (Adapted from Korean J Anesthesiol 2016; 69: 480-6. Fig. 2).

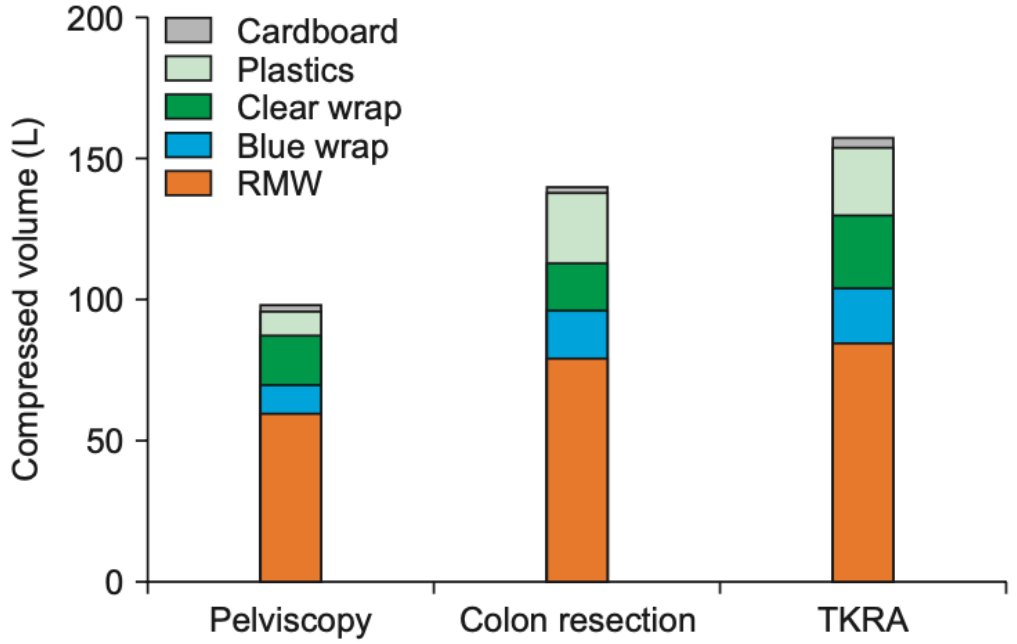
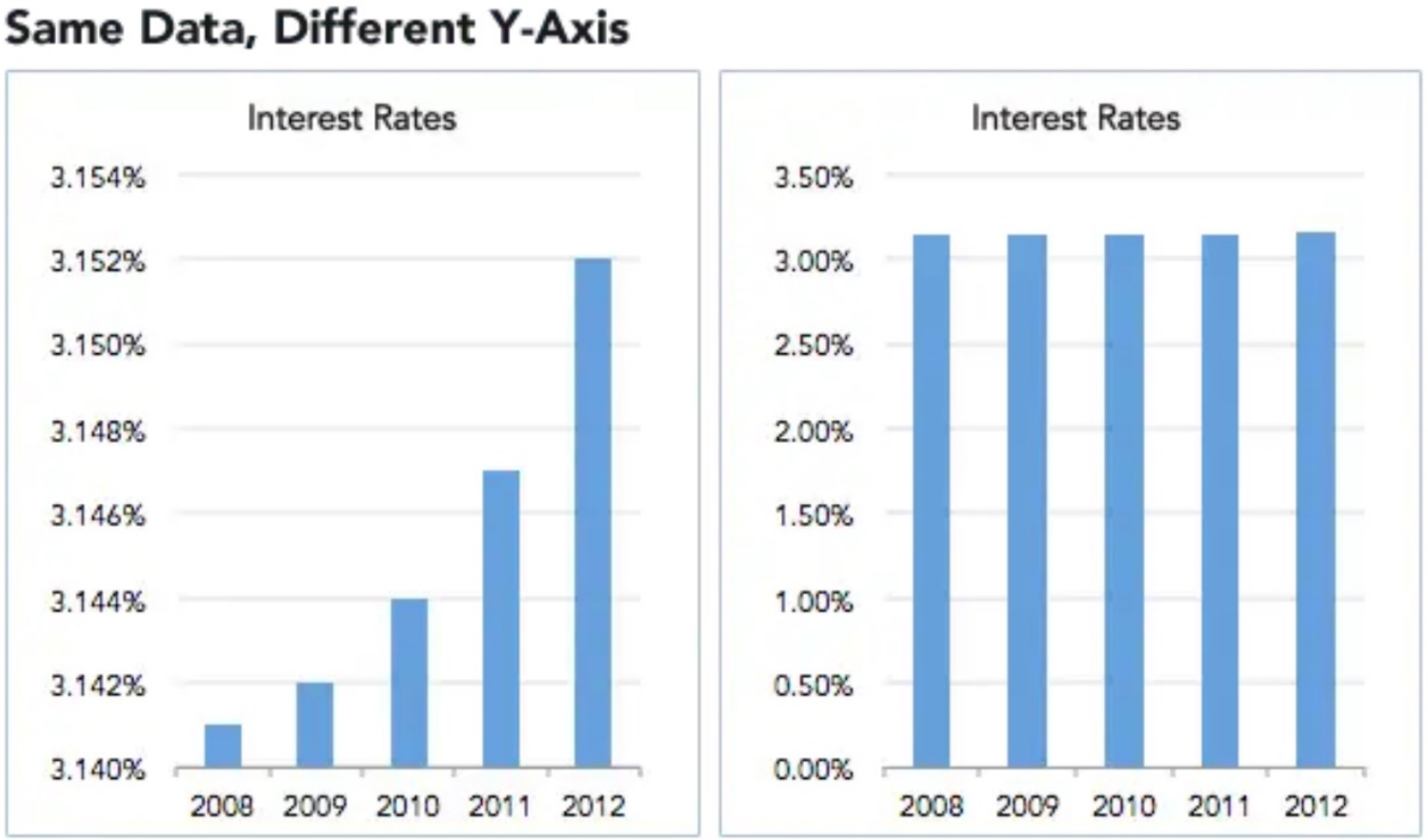
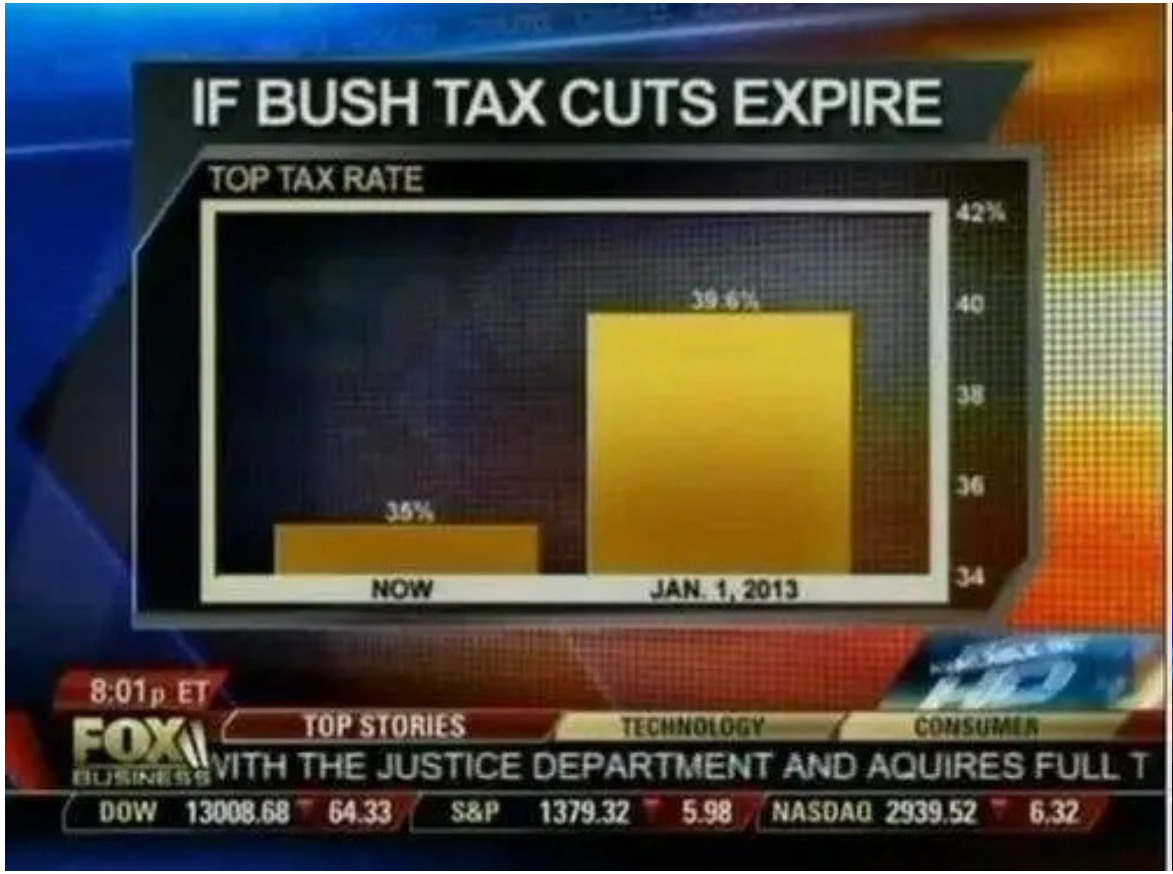


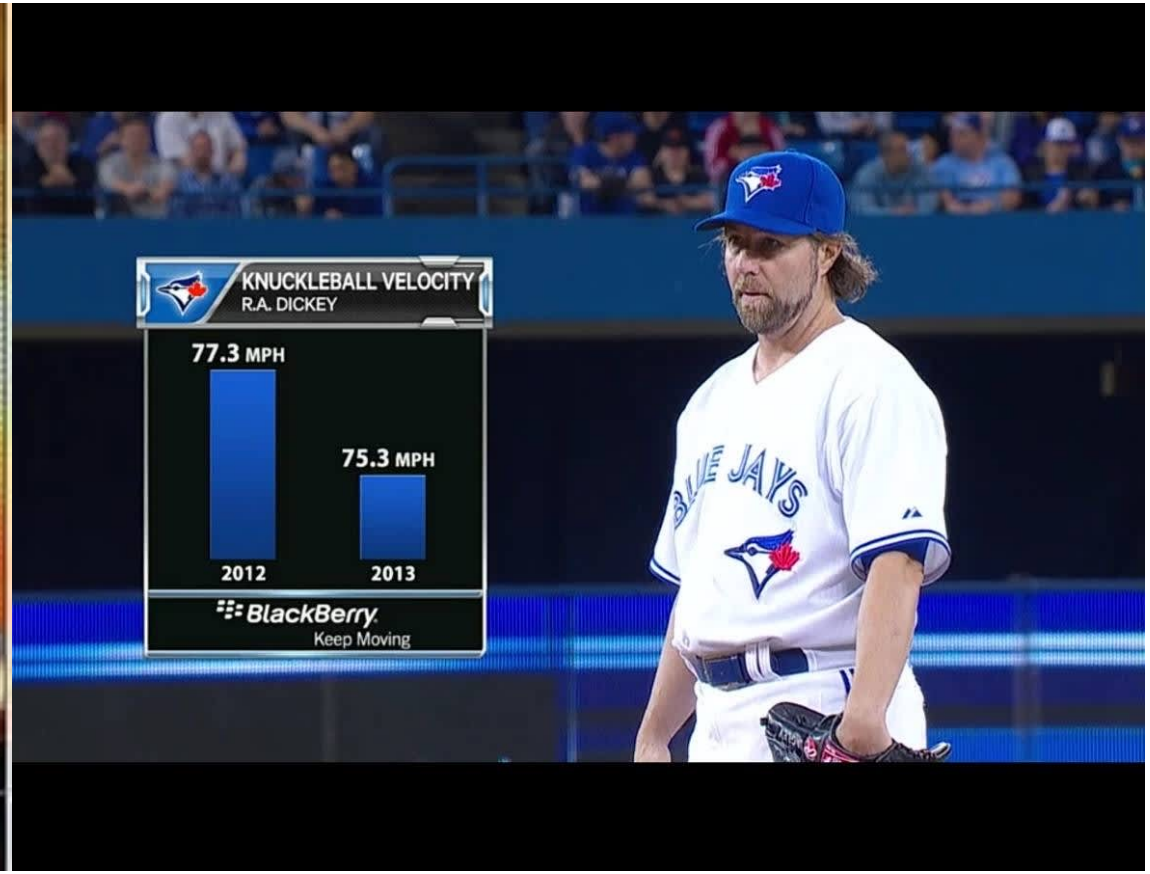
Fig. 4. Stacked bar graph. Compressed volume of each component from the three operations. We checked the compressed volume of each component from the three operations; pelviscopy (with radical vaginal hysterectomy), laparoscopic anterior resection of the colon, and TKRA. TKRA: total knee replacement arthroplasty, RMW: regulated medical waste (Adapted from Korean J Anesthesiol 2017; 70: 100-4).

What could go wrong with a simple bar chart?



What could go wrong with a simple bar chart?





Pie Chart

A pie chart, which is used to represent nominal data (data classified in different categories), visually represents a distribution of categories

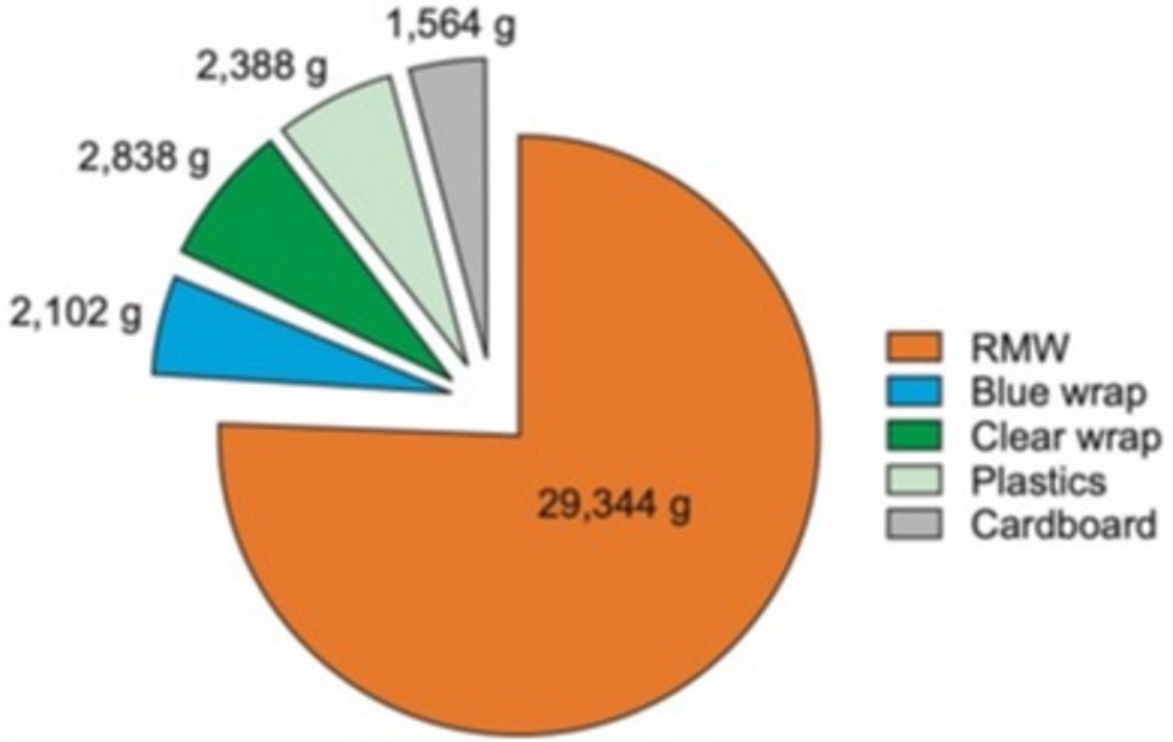


Fig. 5. Pie chart. Total weight of each component from the three operations. RMW: regulated medical waste (Adapted from Korean J Anesthesiol 2017; 70: 100-4).

Pie Chart

A pie chart, which is used to represent nominal data (data classified in different categories), visually re
distributi

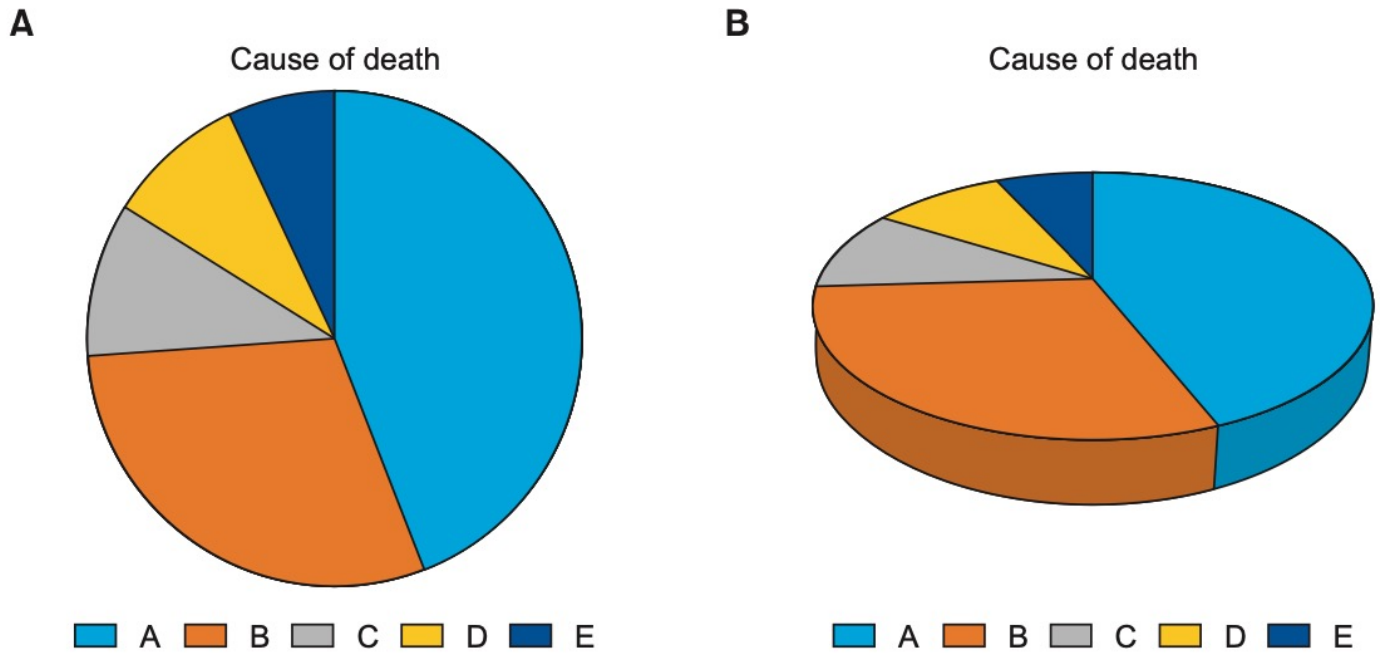
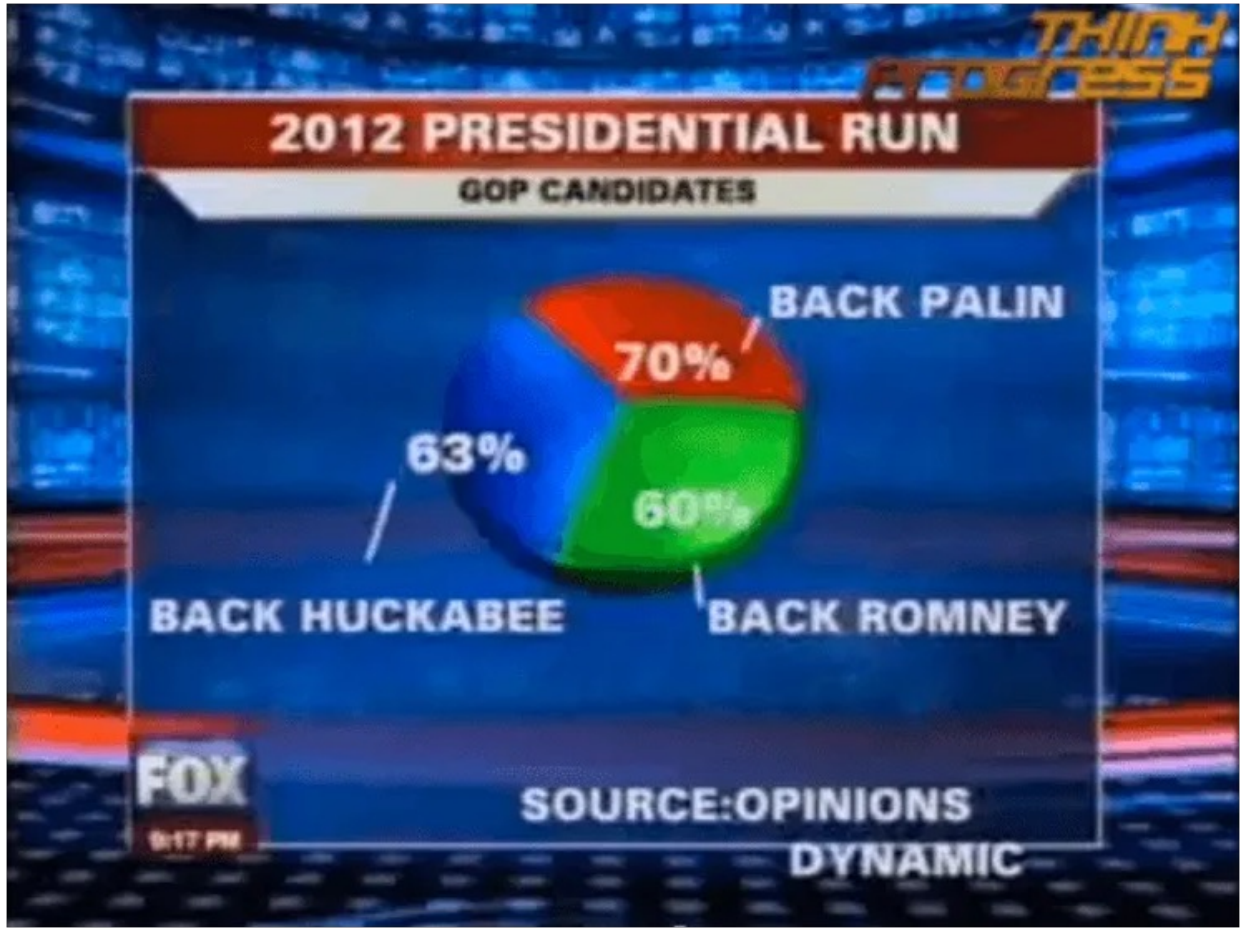


Fig. 8. Simple pie chart (A) versus 3D pie chart (B). In the 3D pie chart, slices at the rear of the chart appear smaller than those at the front because of the false perspective.

What could go wrong with a simple pie chart?



Line Plot

- It is useful for representing time-series data such as monthly precipitation and yearly unemployment rates
- It is used to study variables that are observed over time
- Line graphs are especially useful for studying patterns and trends across data

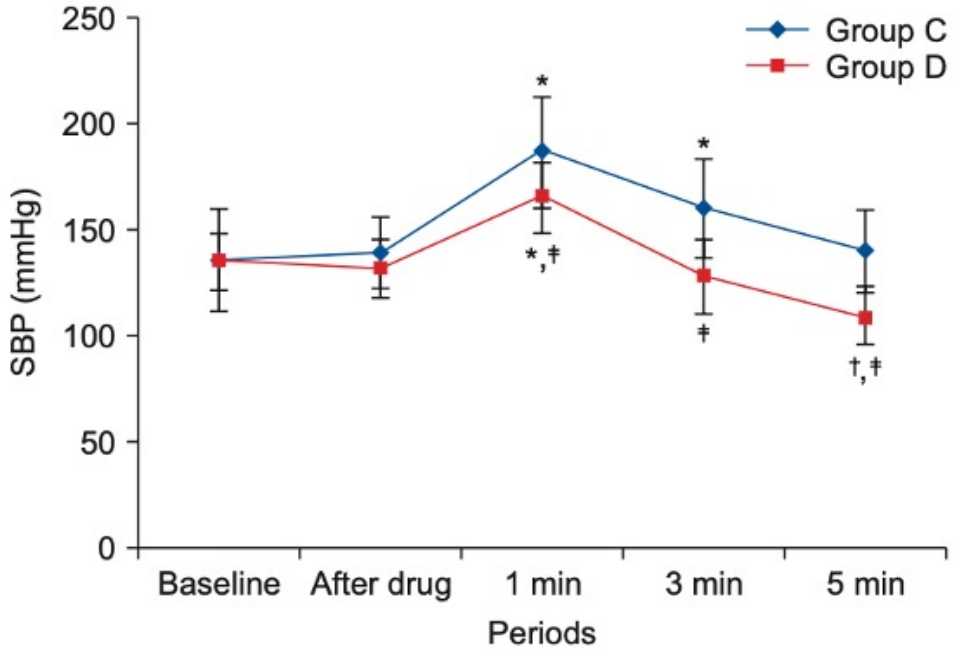


Fig. 1. Line graph with whiskers. Changes in systolic blood pressure (SBP) in the two groups. Group C: normal saline, Group D: dexmedetomidine. *P < 0.05 indicates a significant increase in each group, compared with the baseline values. †P < 0.05 indicates a significant decrease noted in Group D, compared with the baseline values. ‡P < 0.05 indicates a significant difference between the groups (Adapted from Korean J Anesthesiol 2017; 70: 39-45).

Line Plot

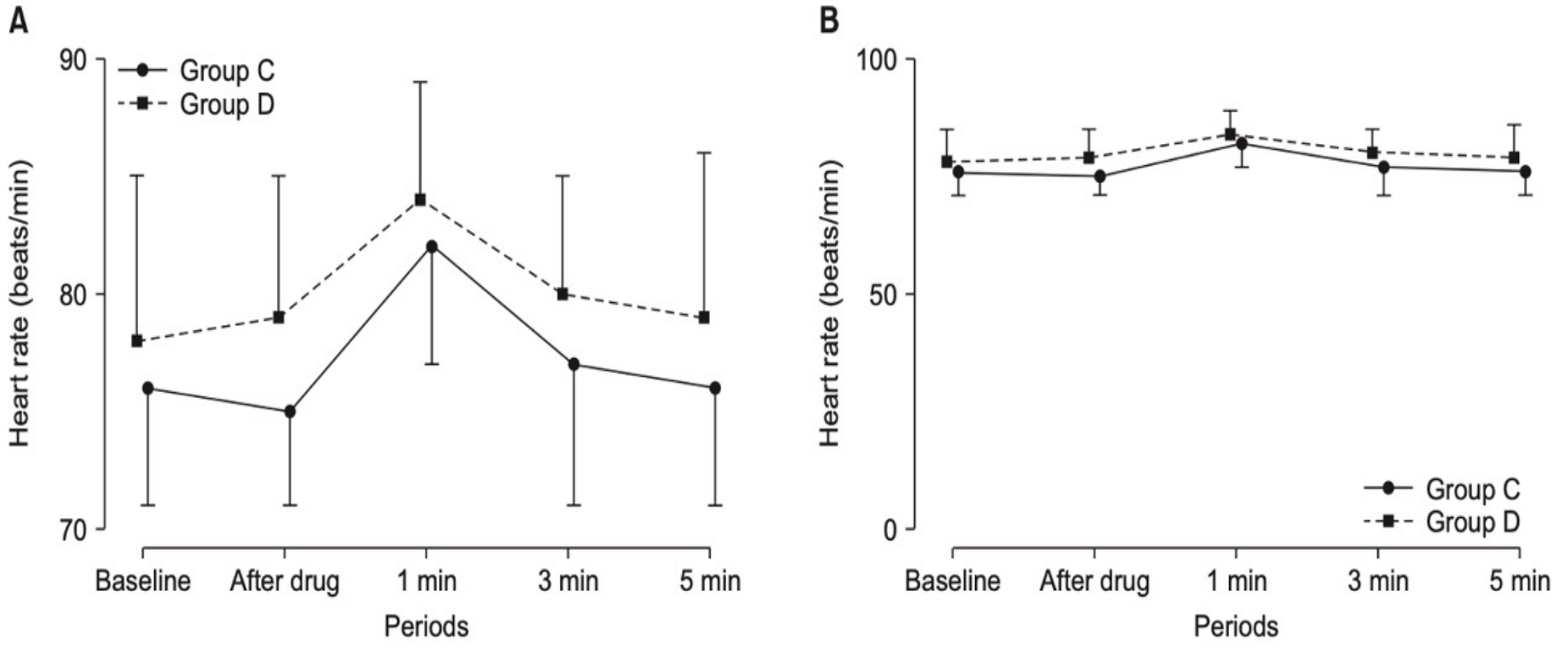
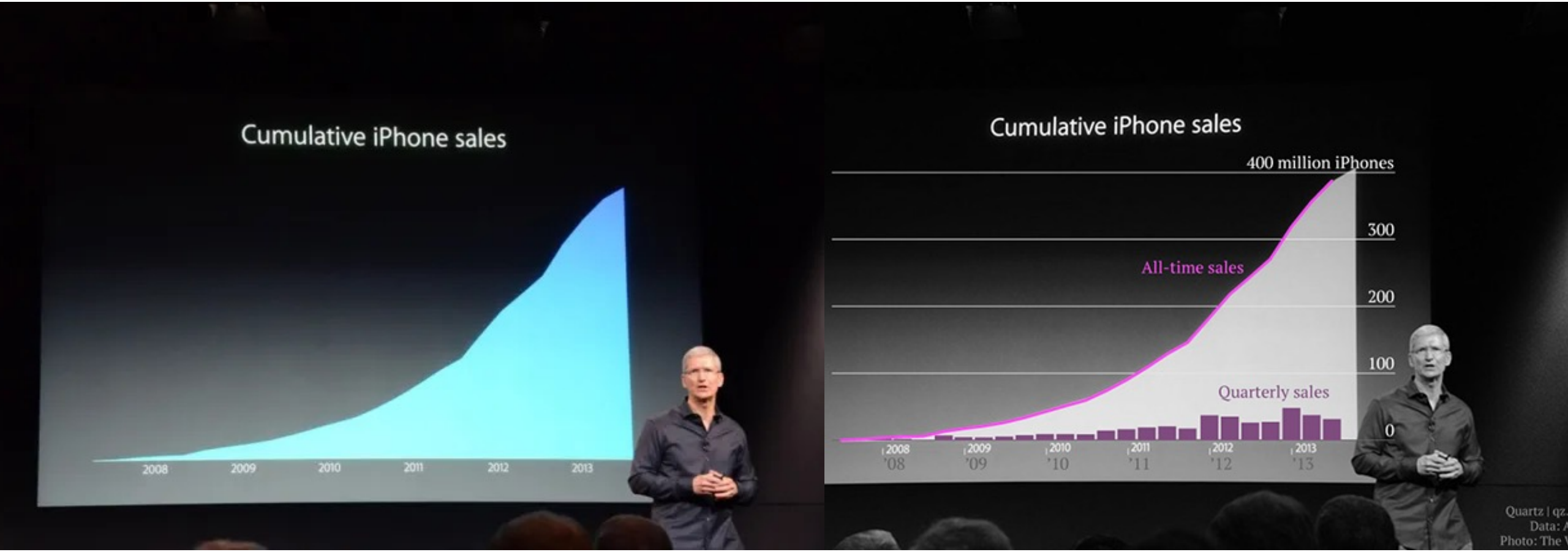


Fig. 10. An example of misleading graphs. Both plots use the same data set. (A) The readers might be thinking that the heart rates at 1 min are higher than others. (B) The heart rates are very stable during the study. The readers should check the scale of the y-axis and baseline values when they look at the graphs.

What could go wrong with a simple line plot?



https://www.techjunkie.com/tim-cook-trying-prove-meaningless-chart/

Waterfall Chart

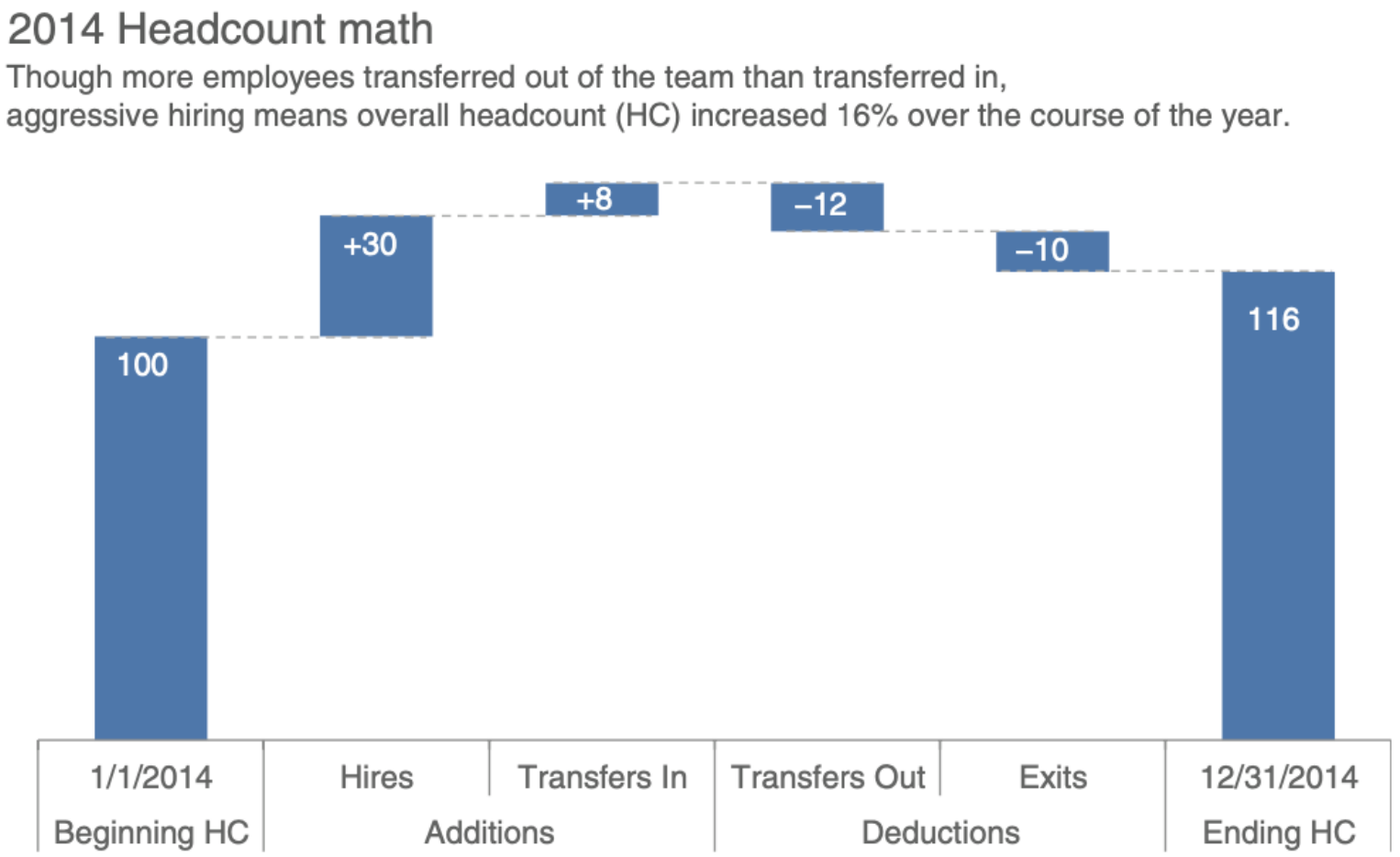
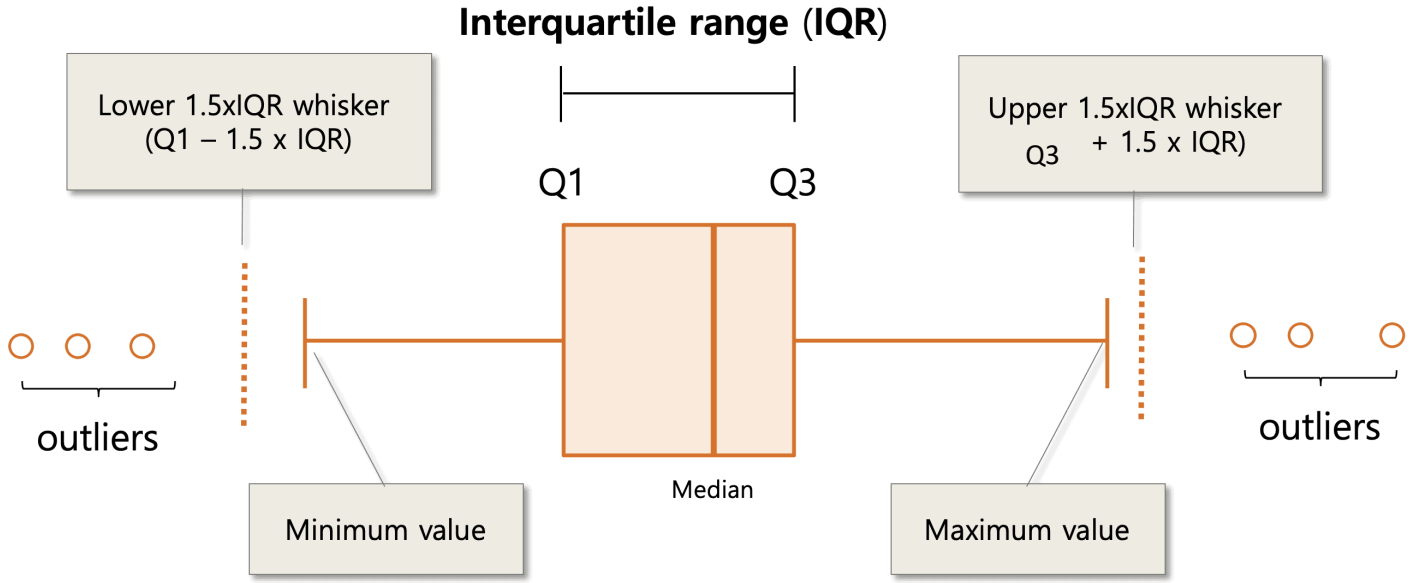


FIGURE 2.17 Waterfall chart

Box and Whisker Chart

- does not make any assumptions about the underlying statistical distribution
- represents variations in samples of a population
- Therefore, c data



https://help.ezbiocloud.net/box-plot/

Box and Whisker Chart

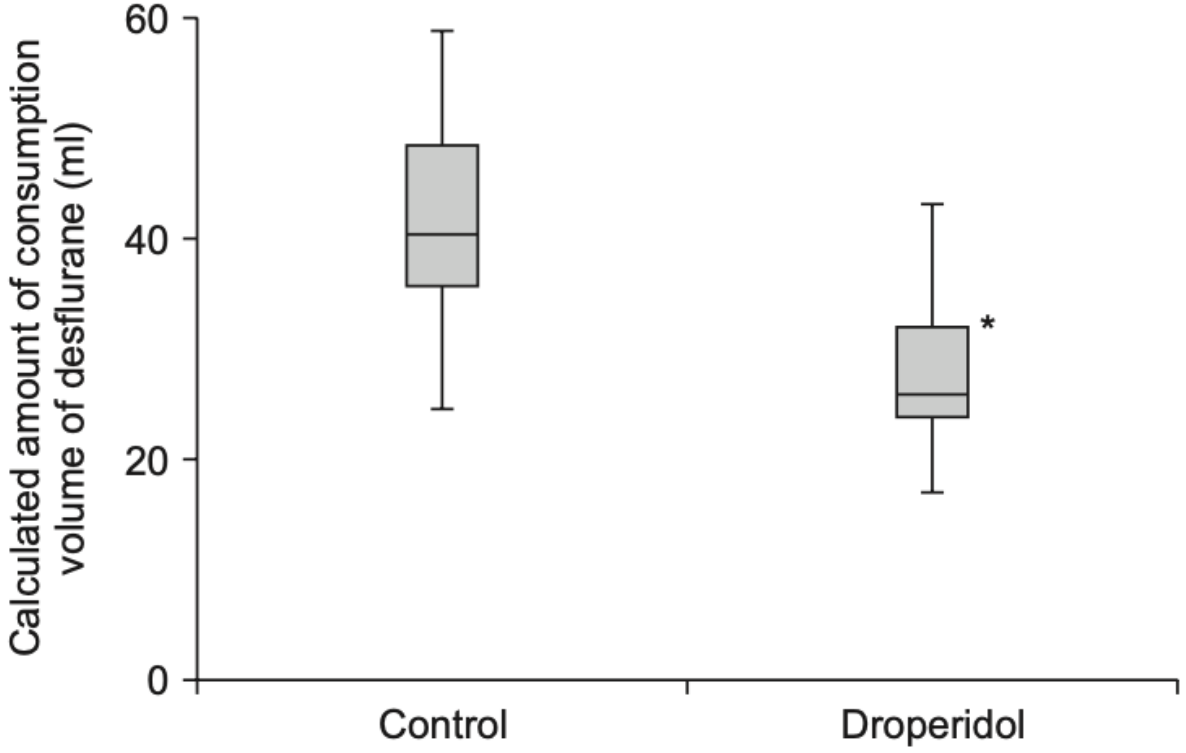


Fig. 7. Box graph with whiskers. Calculated volume of desflurane consumed during the observation period. The groups differed significantly. Data are expressed as median, minimum, first interquartile, third interquartile, and maximum values. *P < 0.05 (Adapted from Korean J Anesthesiol 2017; 70: 27-32).

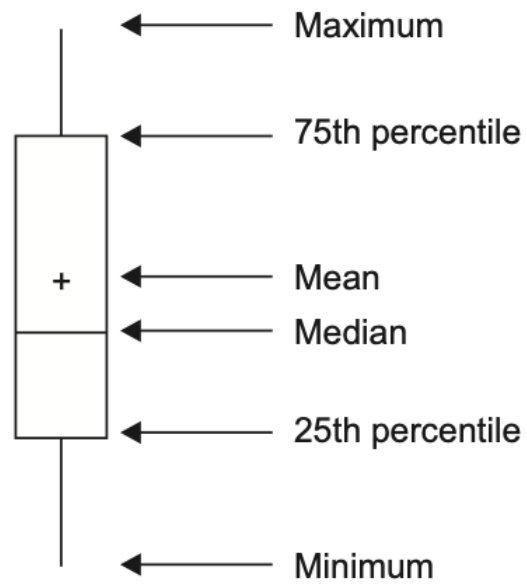


Fig. 6. Box graph with whiskers. This graph is a standardized way of displaying the distribution of data based on the five-number summary; minimum, first quartile, median, third quartile, and maximum. The central rectangle represents from the first quartile to the third quartile (the interquartile range [IQR]). A segment inside the rectangle shows the median and "whiskers" above and below the box show the locations of the minimum and maximum.

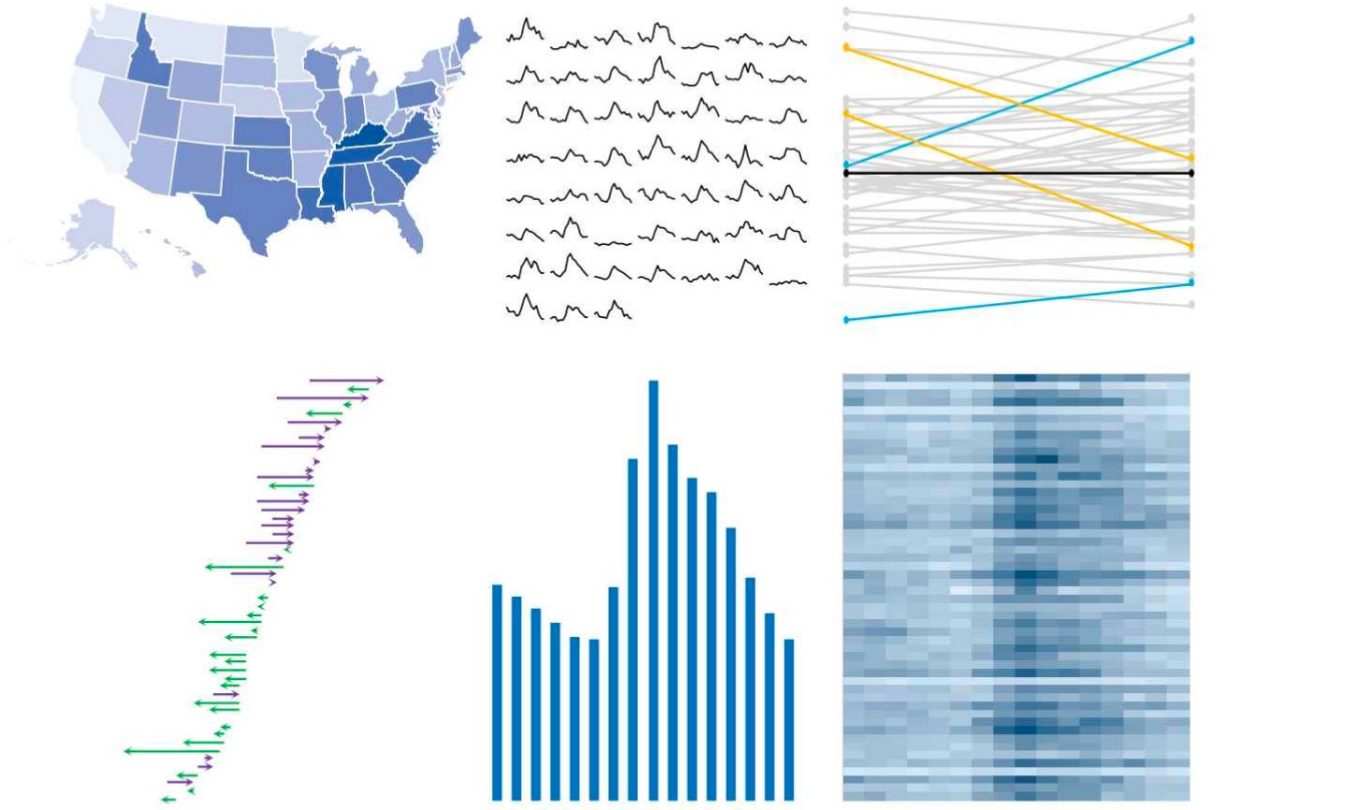
Data Presentation Techniques

Table 3. Types of Charts depending on the Method of Analysis of the Data

Analysis	Subgroup	Number of variables	Type
Comparison	Among items	Two per items	Variable width column chart
		One per item	Bar/column chart
	Over time	Many periods	Circular area/line chart
		Few periods	Column/line chart
Relationship		Two	Scatter chart
		Three	Bubble chart
Distribution		Single	Column/line histogram
		Two	Scatter chart
		Three	Three-dimensional area chart
Comparison	Changing over time	Only relative differences matter	Stacked 100% column chart
		Relative and absolute differences matter	Stacked column chart
	Static	Simple share of total	Pie chart
		Accumulation	Waterfall chart
		Components of components	Stacked 100% column chart with subcomponents

Data Presentation Techniques

- How should I present the data?



Each of these six charts visualizes the same data: The share of people earning minimum wage or less in each state.

Better Data Visualizations A Guide for Scholars, Researchers, and Wonks (Jonathan Schwabish)

Data Presentation Techniques

- How should I present the data?

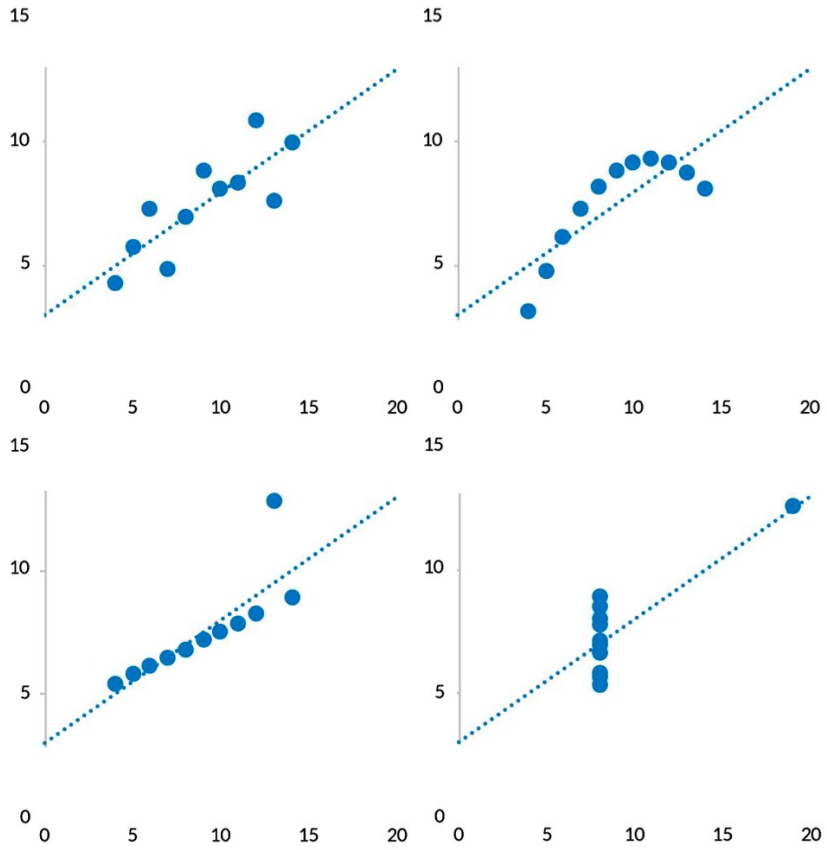
Data set		1	1	2	2	3	3	4	4
Variable		x	y	x	y	x	y	x	y
Obs. No.	1 :	10	8.0	10	9.1	10	7.5	8	6.6
	2 :	8	7.0	8	8.1	8	6.8	8	5.8
	3 :	13	7.6	13	8.7	13	12.7	8	7.7
	4 :	9	8.8	9	8.8	9	7.1	8	8.8
	5 :	11	8.3	11	9.3	11	7.8	8	8.5
	6 :	14	10.0	14	8.1	14	8.8	8	7.0
	7 :	6	7.2	6	6.1	6	6.1	8	5.3
	8 :	4	4.3	4	3.1	4	5.4	19	12.5
	9 :	12	10.8	12	9.1	12	8.2	8	5.6
	10 :	7	4.8	7	7.3	7	6.4	8	7.9
	11 :	5	5.7	5	4.7	5	5.7	8	6.9
Mean		9.0	7.5	9.0	7.5	9.0	7.5	9.0	7.5
Variance		11.0	4.1	11.0	4.1	11.0	4.1	11.0	4.1
Correlation		0.816		0.816		0.816		0.817	
Regression line		y = 3 + 0.5x		y = 3 + 0.5x		y = 3 + 0.5x		y = 3 + 0.5x	

Source: Francis Anscombe

Known as Anscombe's Quartet, this example demonstrates how difficult it is for us to pull out basic patterns and summary statistics.

Data Presentation Techniques

- How should I present the data?



The data visualization representation of Anscombe's Quartet. Notice how much easier it is to see the positive relationship between the two variables, the curvature in the pattern in the top-right graph, and the outliers in the bottom two graphs.

Source: Francis Anscombe (1973).

Data Presentation Techniques

- How should I present the data?

Table 1. Our sales grew to $600 million this year

	Q1	Q2	Q3	Q4
Bob	26	35	72	84
Ellie	22	15	61	35
Gerrie	19	20	71	55
Jack	22	95	13	64
Jon	83	62	46	48
Karen	30	65	98	82
Ken	38	28	45	71
Lauren	98	81	41	63
Steve	16	50	23	41
Valerie	46	24	30	57
Total	**$400**	**$475**	**$500**	**$600**

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Data Presentation Techniques

- How should I present the data?

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Total	**$400**	**$475**	**$500**	**$600**

Table 1. Our sales grew to $600 million this year

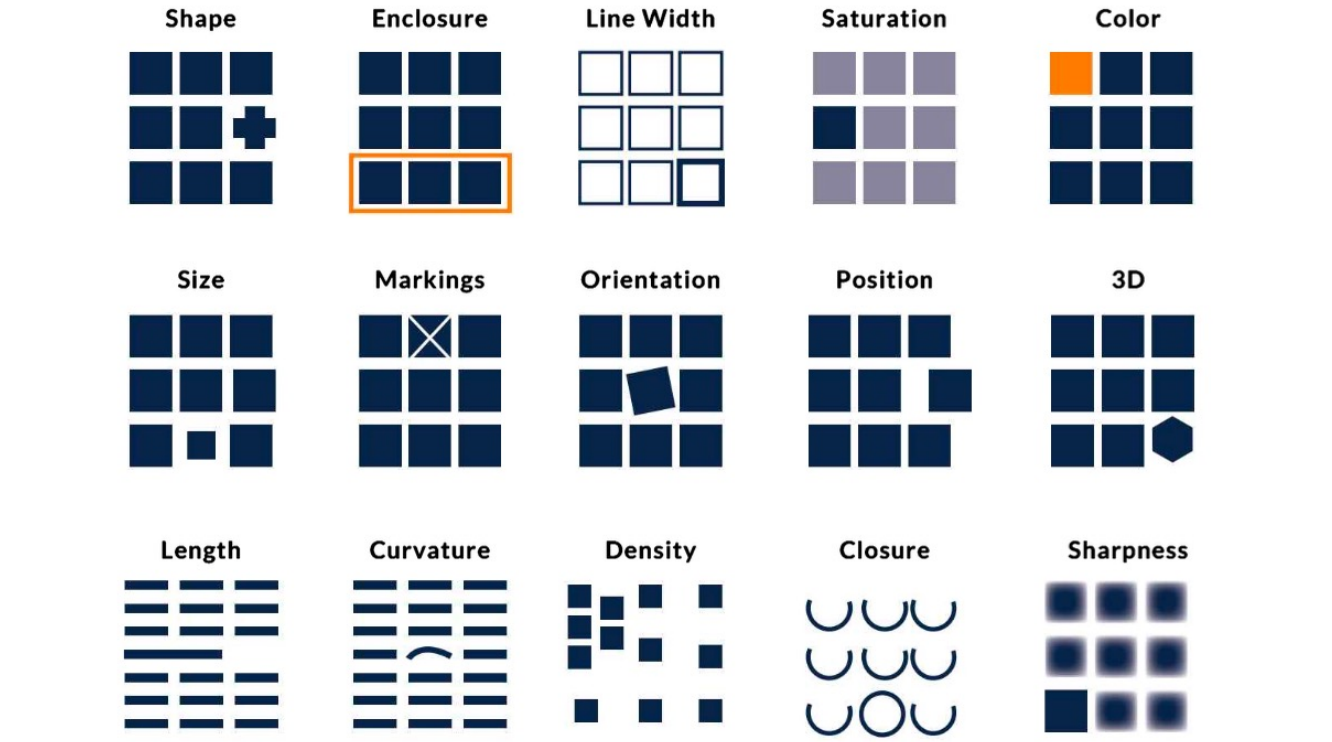
	Q1	Q2	Q3	Q4
Bob	26	35	72	**84**
Ellie	22	15	61	35
Gerrie	19	20	71	55
Jack	22	**95**	13	64
Jon	83	62	46	48
Karen	30	65	**98**	82
Ken	38	28	45	71
Lauren	**98**	81	41	63
Steve	16	50	23	41
Valerie	46	24	30	57
Total	**$400**	**$475**	**$500**	**$600**

Preattentive attributes here direct our attention to the large numbers immediately.

Better Data Visualizations A Guide for Scholars, Researchers, and Wonks (Jonathan Schwabish)

Data Presentation Techniques

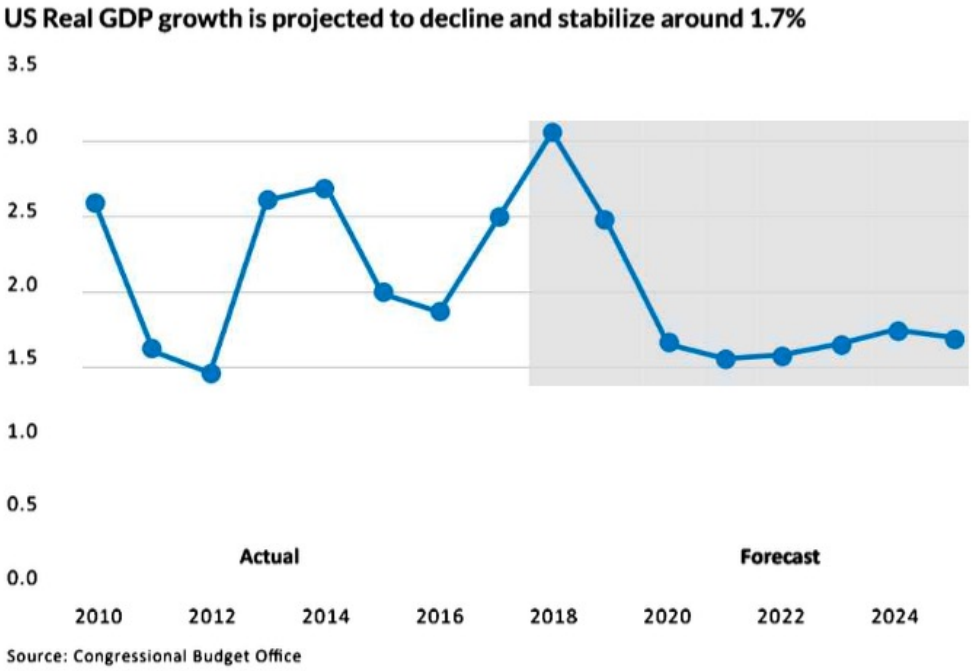
- How should I present the data?

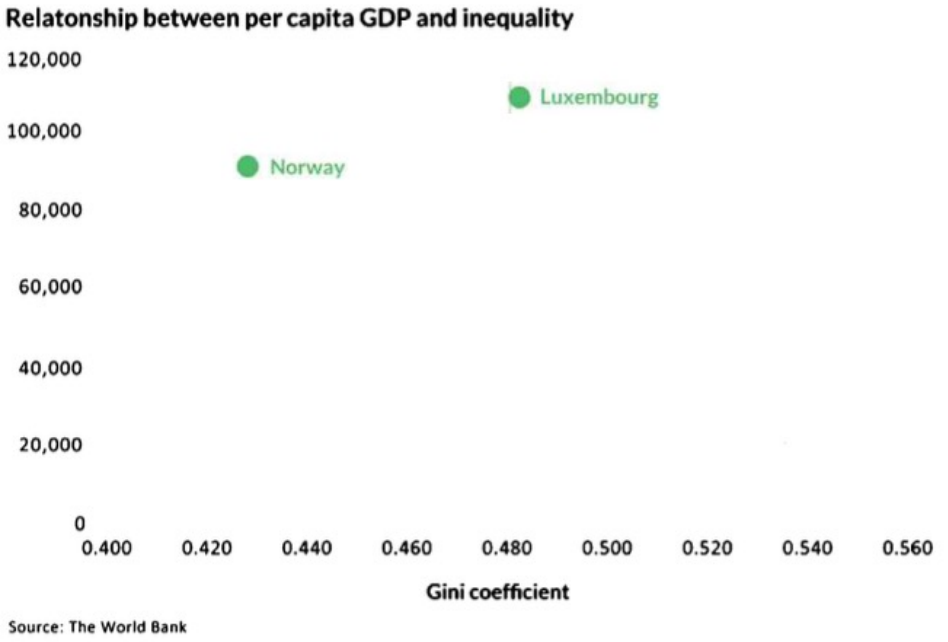


Examples of preattentive attributes that we can use in our visualizations to direct our reader's attention.

Better Data Visualizations A Guide for Scholars, Researchers, and Wonks (Jonathan Schwabish)

Data Presentation Techniques





Better Data Visualizations A Guide for Scholars, Researchers, and Wonks (Jonathan Schwabish)

Data Presentation Techniques

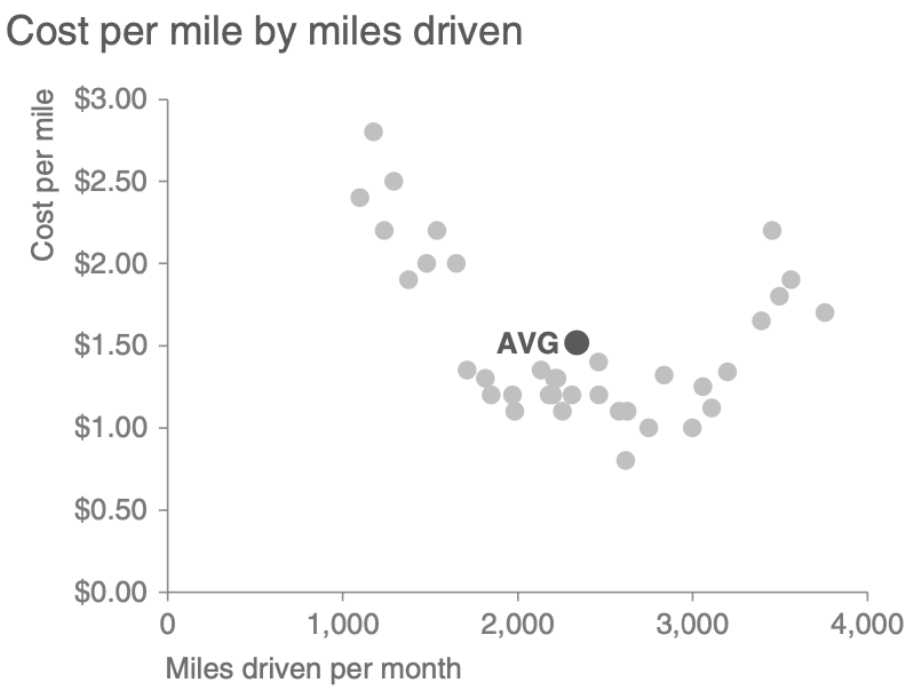


FIGURE 2.6 Scatterplot

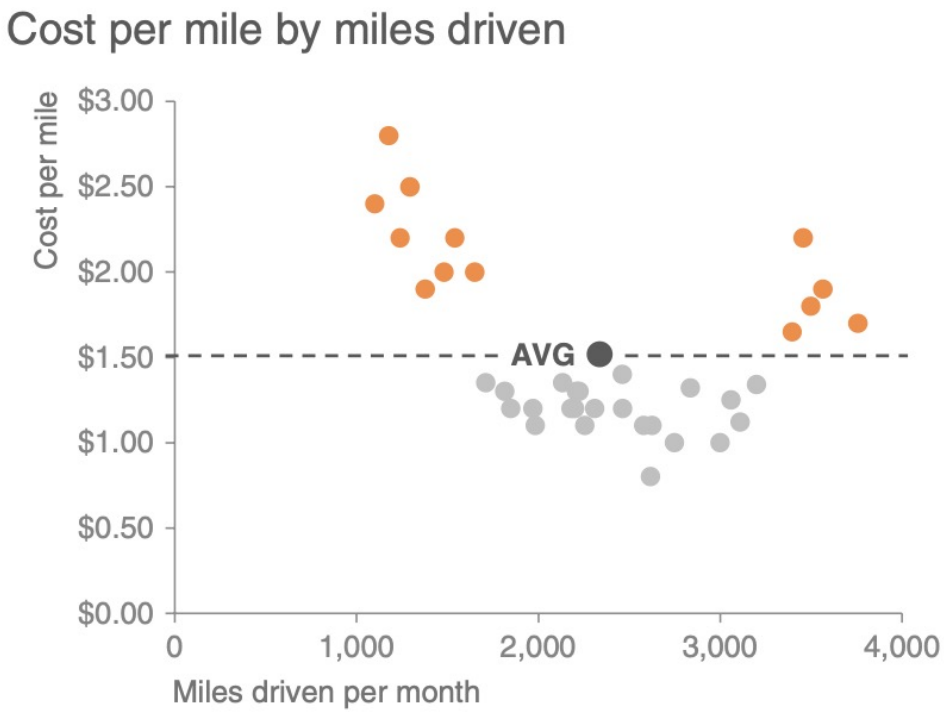


FIGURE 2.7 Modified scatterplot

Data Presentation Techniques

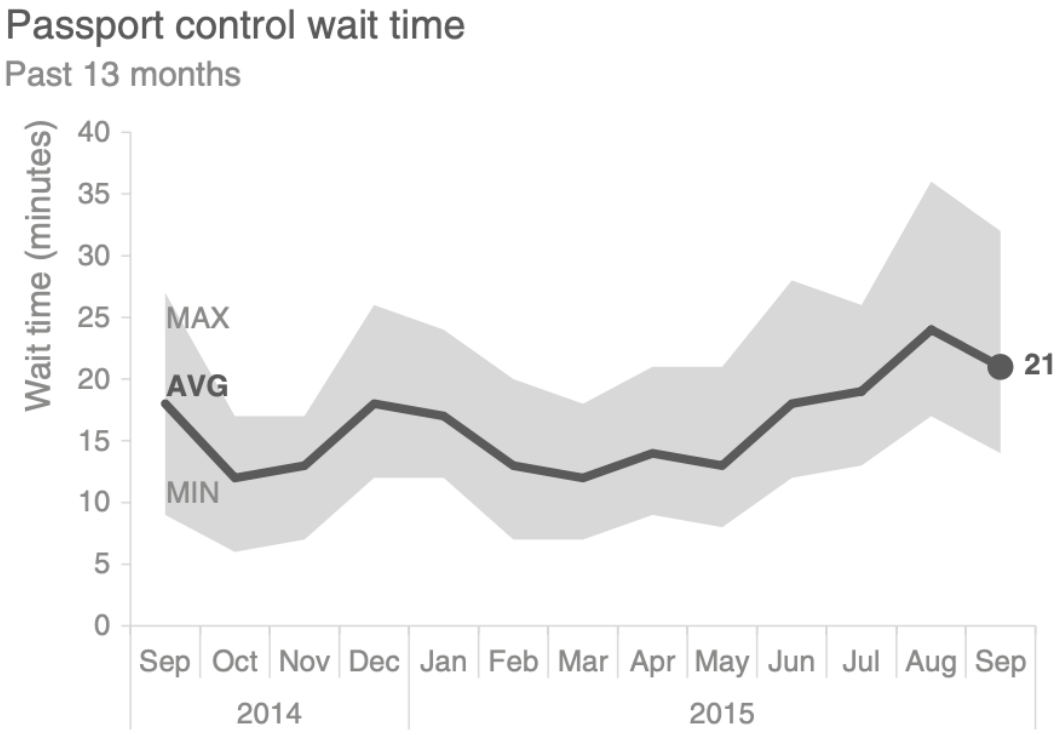


FIGURE 2.9 Showing average within a range in a line graph

Data Presentation Techniques

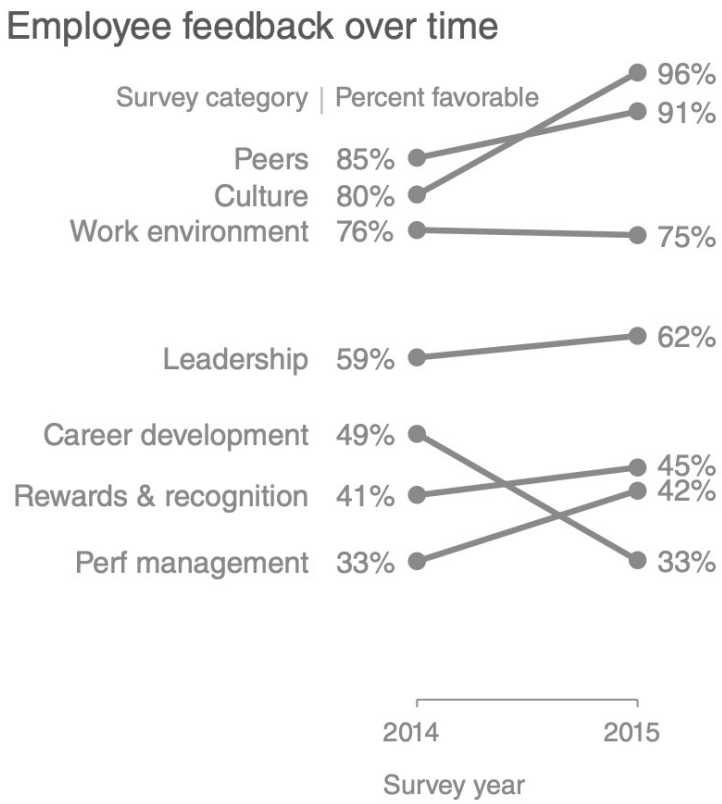


FIGURE 2.10 Slopegraph

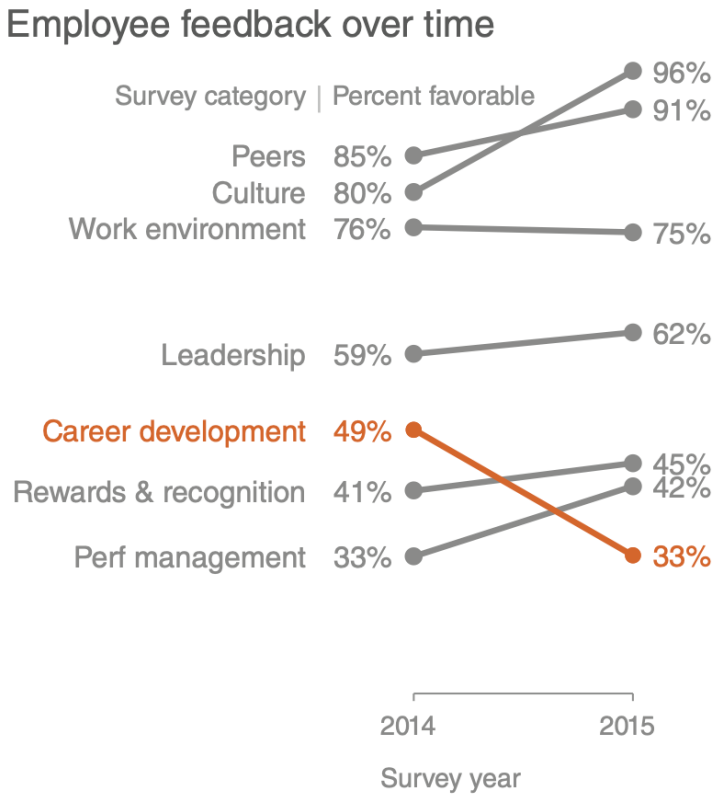


FIGURE 2.11 Modified slopegraph

storytellingwithdata.com/slopegraph-template

Data Presentation Techniques

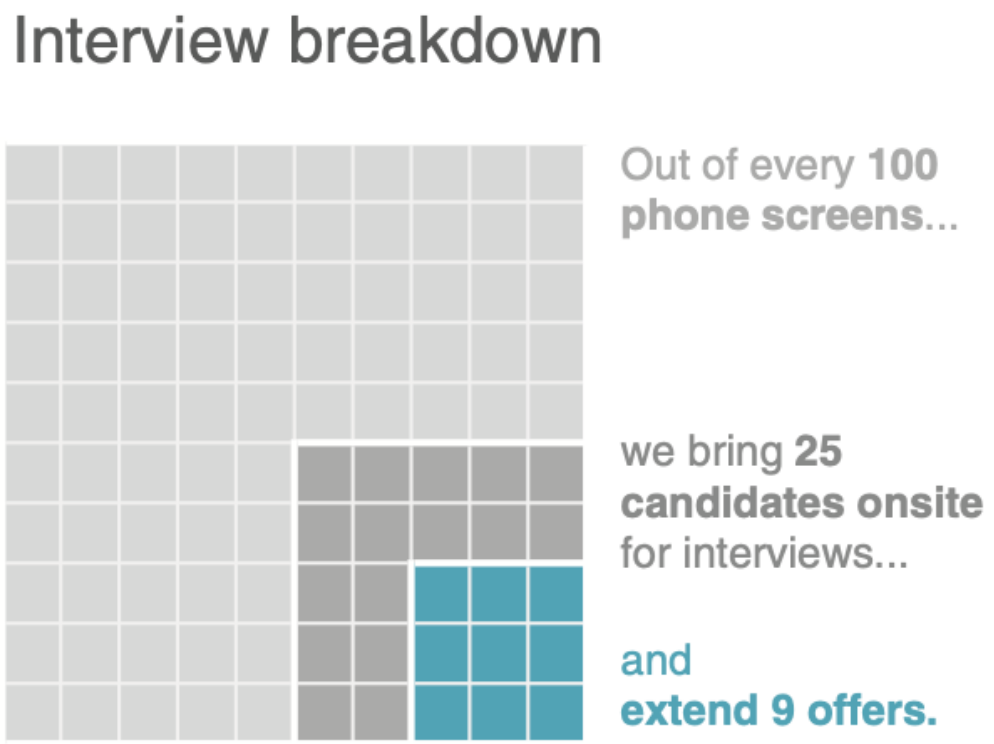


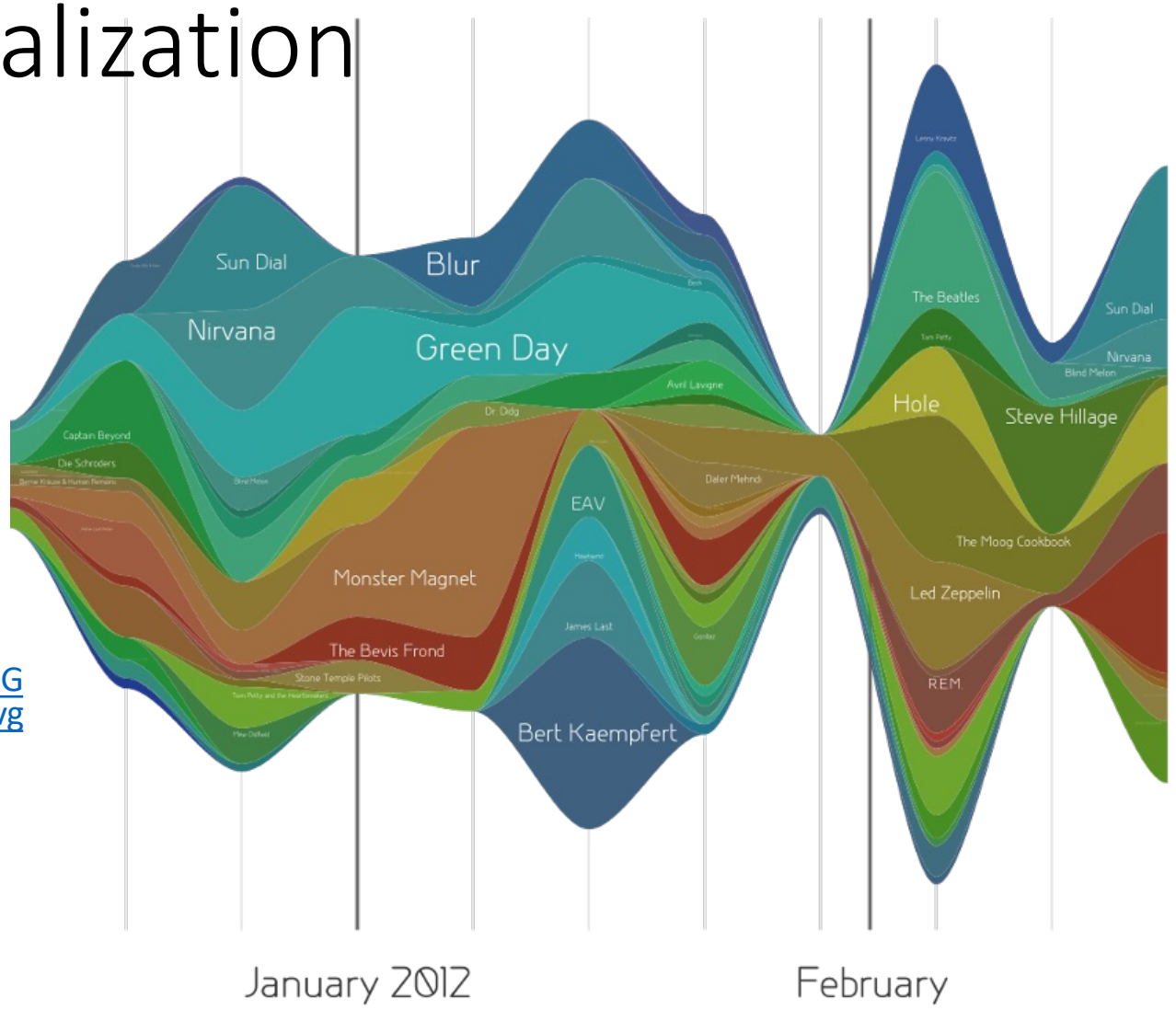
FIGURE 2.20 Square area graph

Multi-dimensional Visualization

- Temporal – Streamgraph
  - A streamgraph showing someone's Last.fm music listening habits over time.

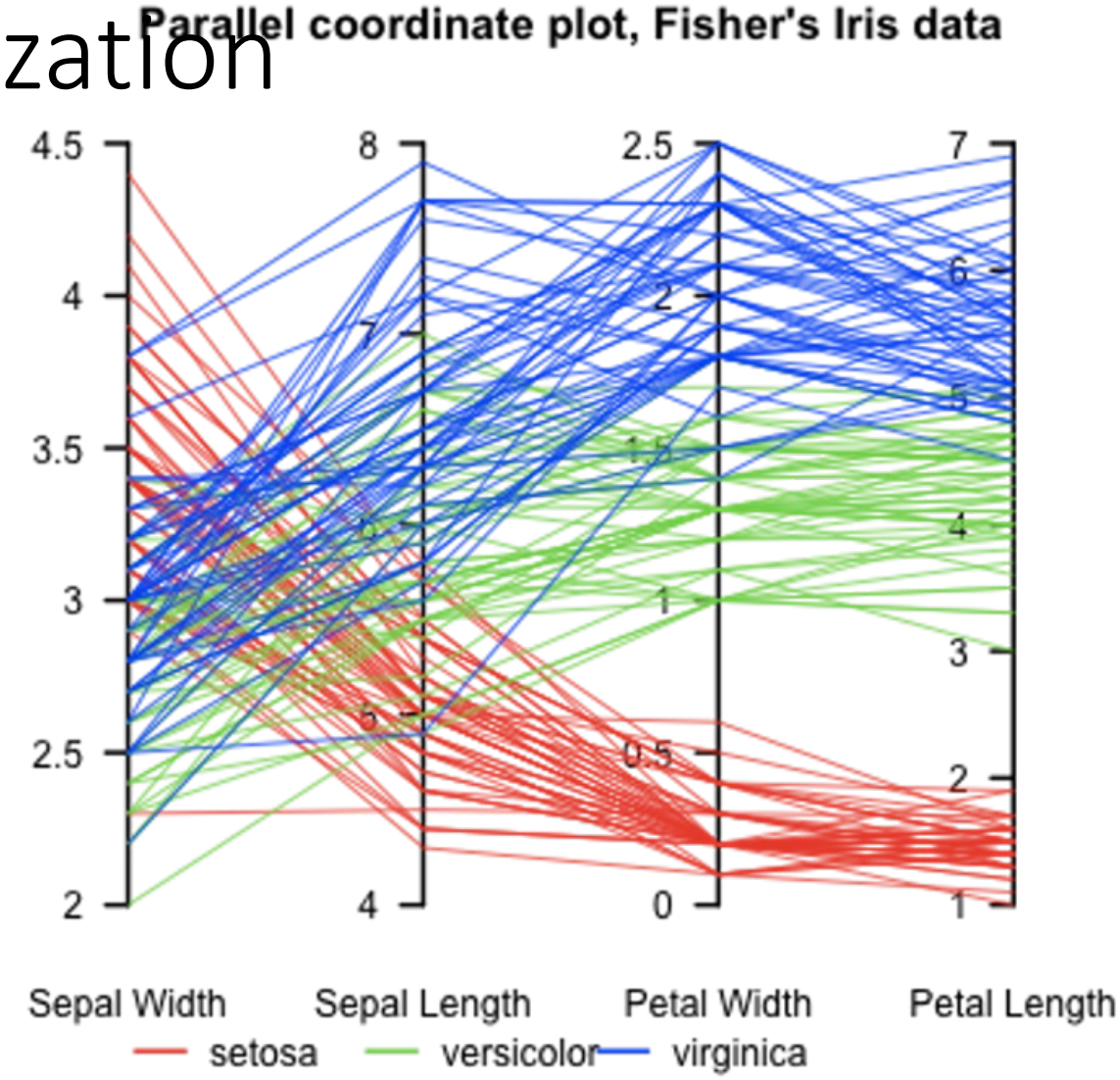
Produced with LastGraph.

https://en.wikipedia.org/wiki/Streamgraph#/media/File:LastGraph_example.svg



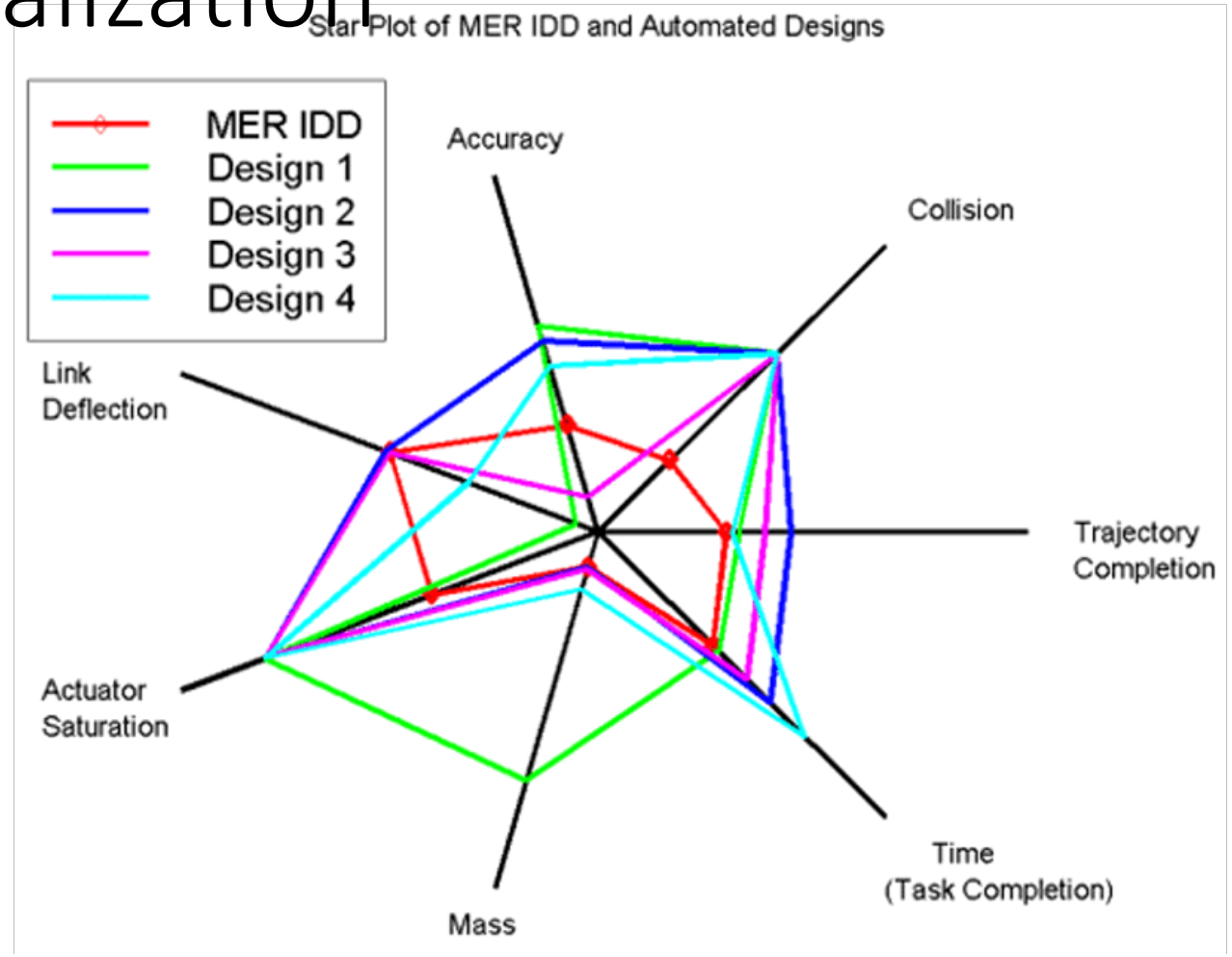
Multi-dimensional Visualization

- Parallel coordinates
  - ParCorFisherIris https://en.wikipedia.org/wiki/File:ParCorFisherIris.png#/media/File:ParCorFisherIris.png
  - Nutrition https://jwatt.org/tmp/exposedata.com/parallel/



Multi-dimensional Visualization

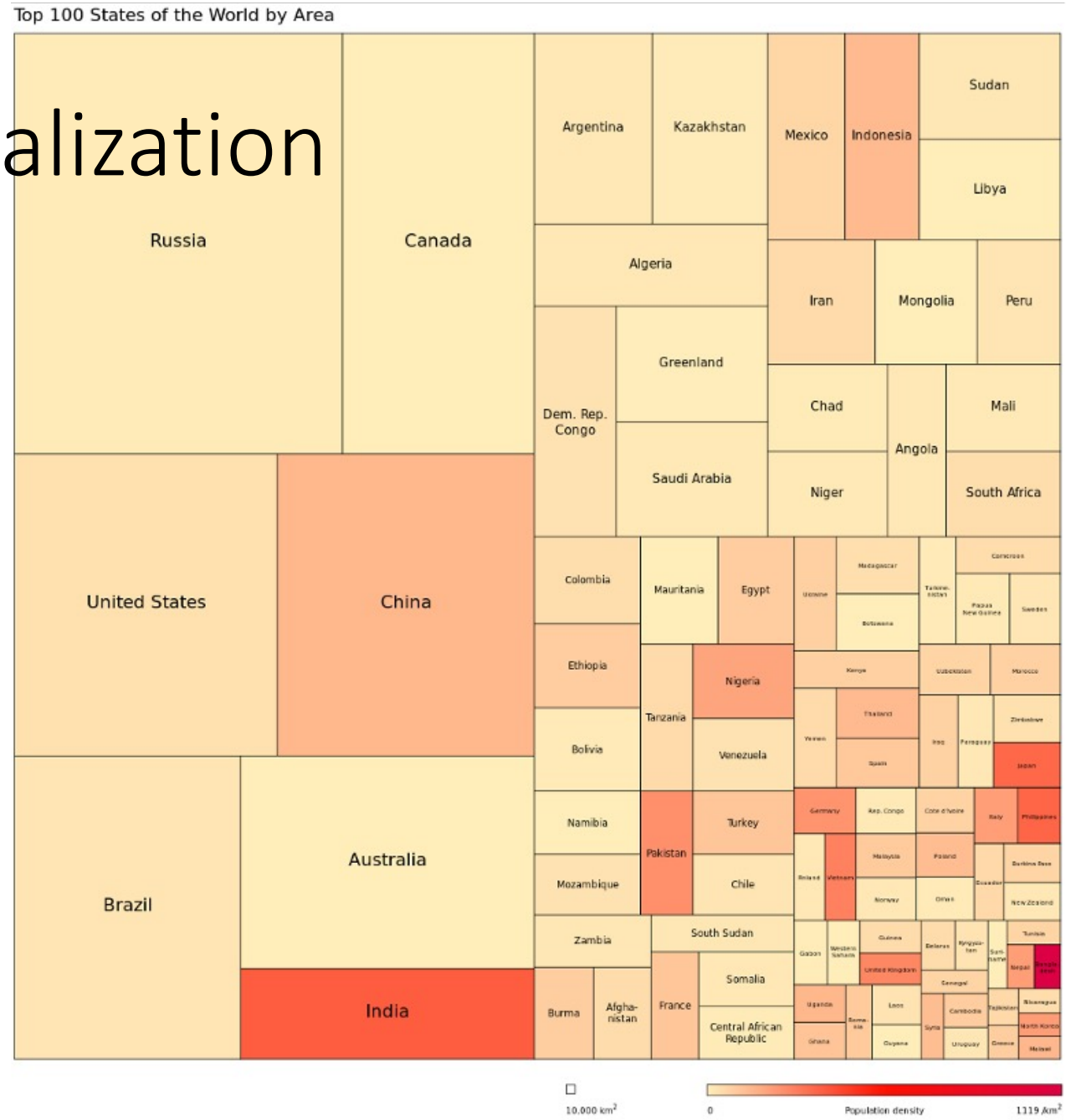
- Star plot
  - NASA Space system design: Desirable is in the middle.
  - The red line represents the handcrafted MER IDD, and is used for comparison and validation of the approach.
  - This graph illustrates the fact that improvements in one metric often come at the expense of others.
  - However, design samples 3 and 4, which were among the last of the 10 actually produced, show the best overall results in the most important metrics.



https://en.wikipedia.org/wiki/File:MER_Star_Plot.gif#/media/File:MER_Star_Plot.gif

Multi-dimensional Visualization

- Treemap
  - The world's top 100 countries sorted by area.
  - Color indicates population density.

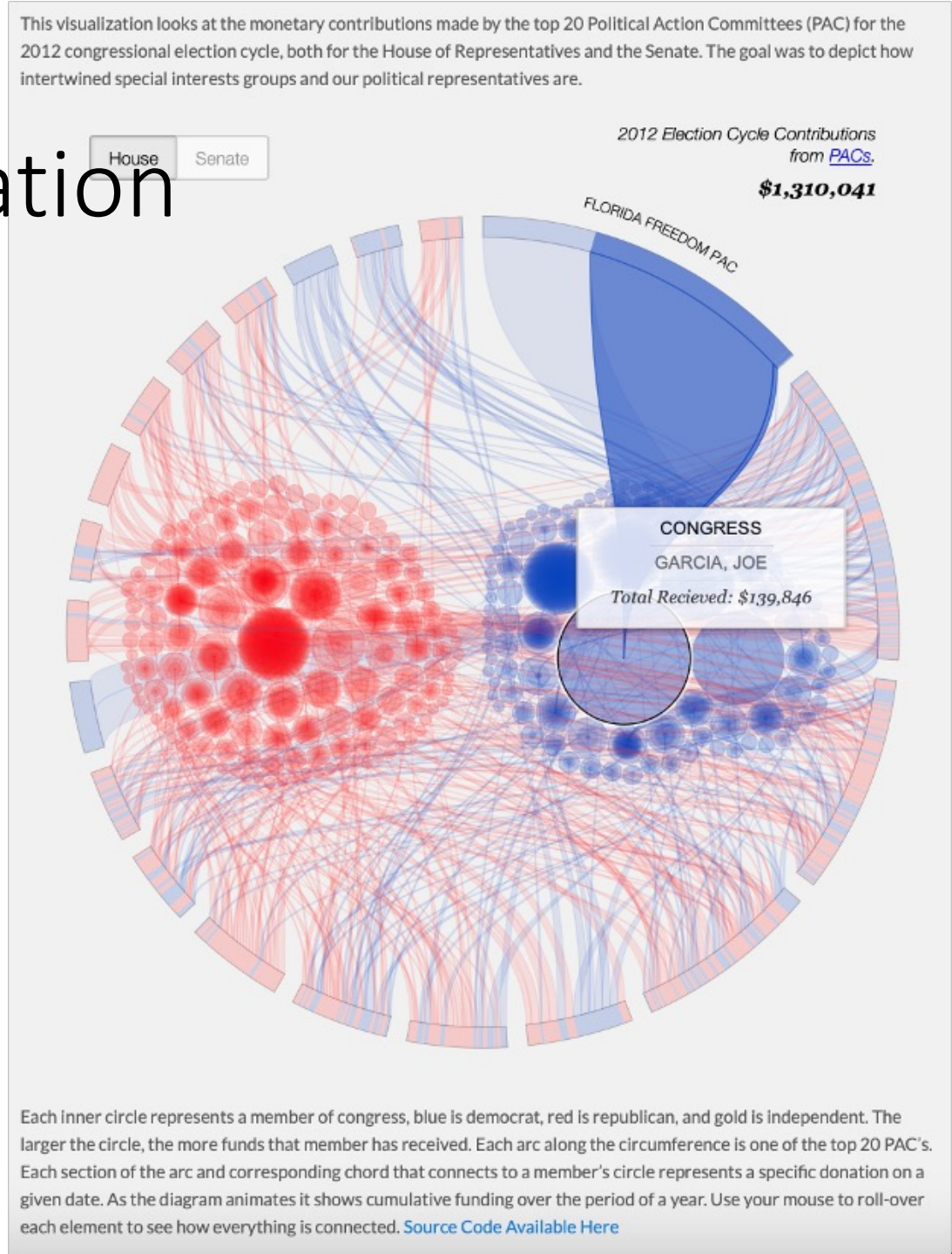


https://commons.wikimedia.org/wiki/File:Top100_states_area_treemap_pop-density.svg#/media/File:Top100_states_area_treemap_pop-density.svg

Multi-dimensional Visualization

- Chore Diagram
  - This visualization looks at the monetary contributions made by the top 20 Political Action Committees (PAC) for the 2012 congressional election cycle, both for the House of Representatives and the Senate.

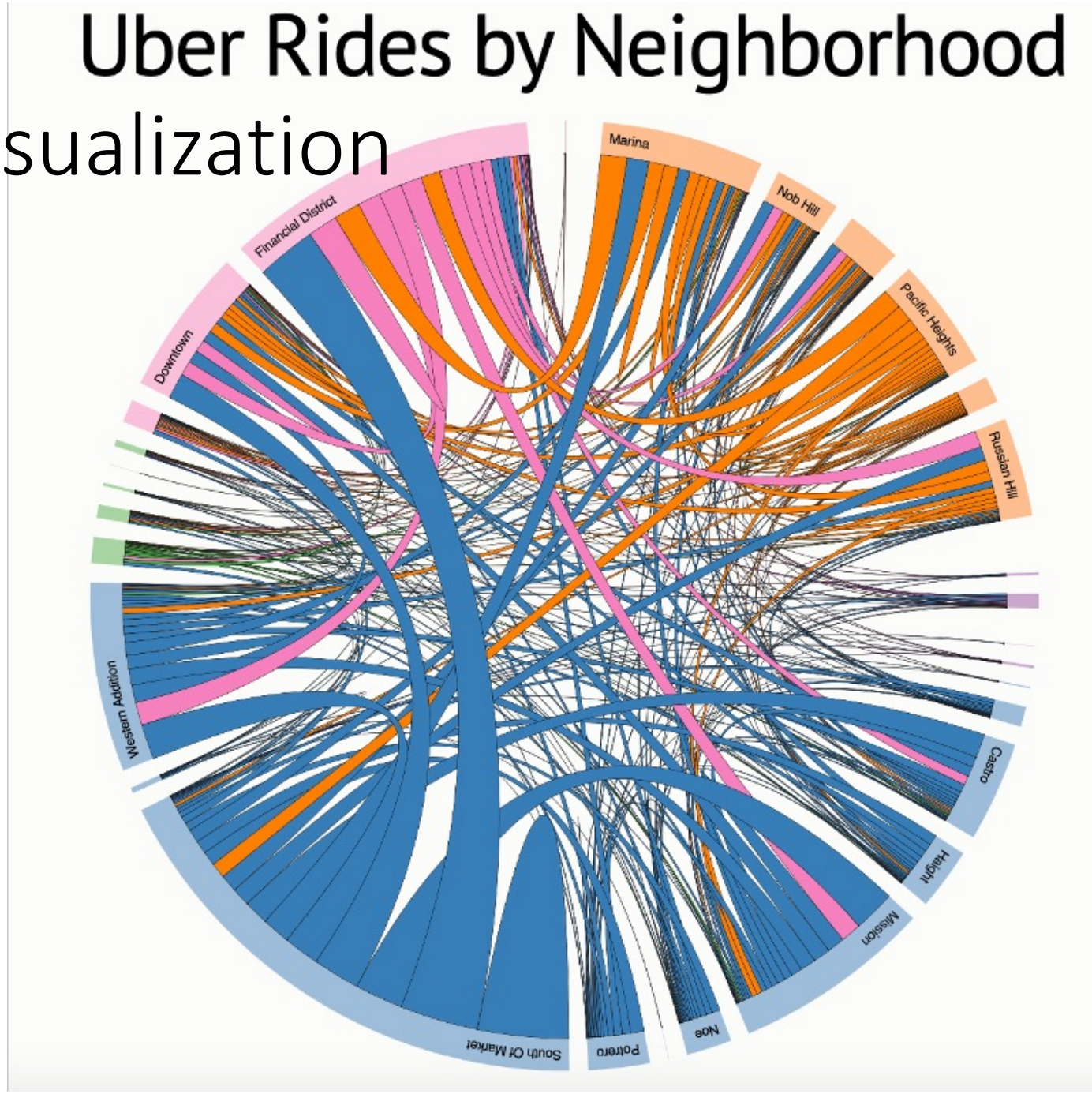
http://www.brightpointinc.com/political_influence/?source=d3js



Multi-dimensional Visualization

- Chore Diagram
  - Mouseover to focus on rides to or from a single neighborhood.
  - The thickness of links between neighborhoods encodes the relative frequency of rides between two neighborhoods: thicker links represent more frequent rides.
  - Links are directed: for example, while 2.2% of rides go from South of Market to Downtown, only 1.2% go in the opposite direction. Links are colored by the more frequent origin.

https://bost.ocks.org/mike/uberdata/

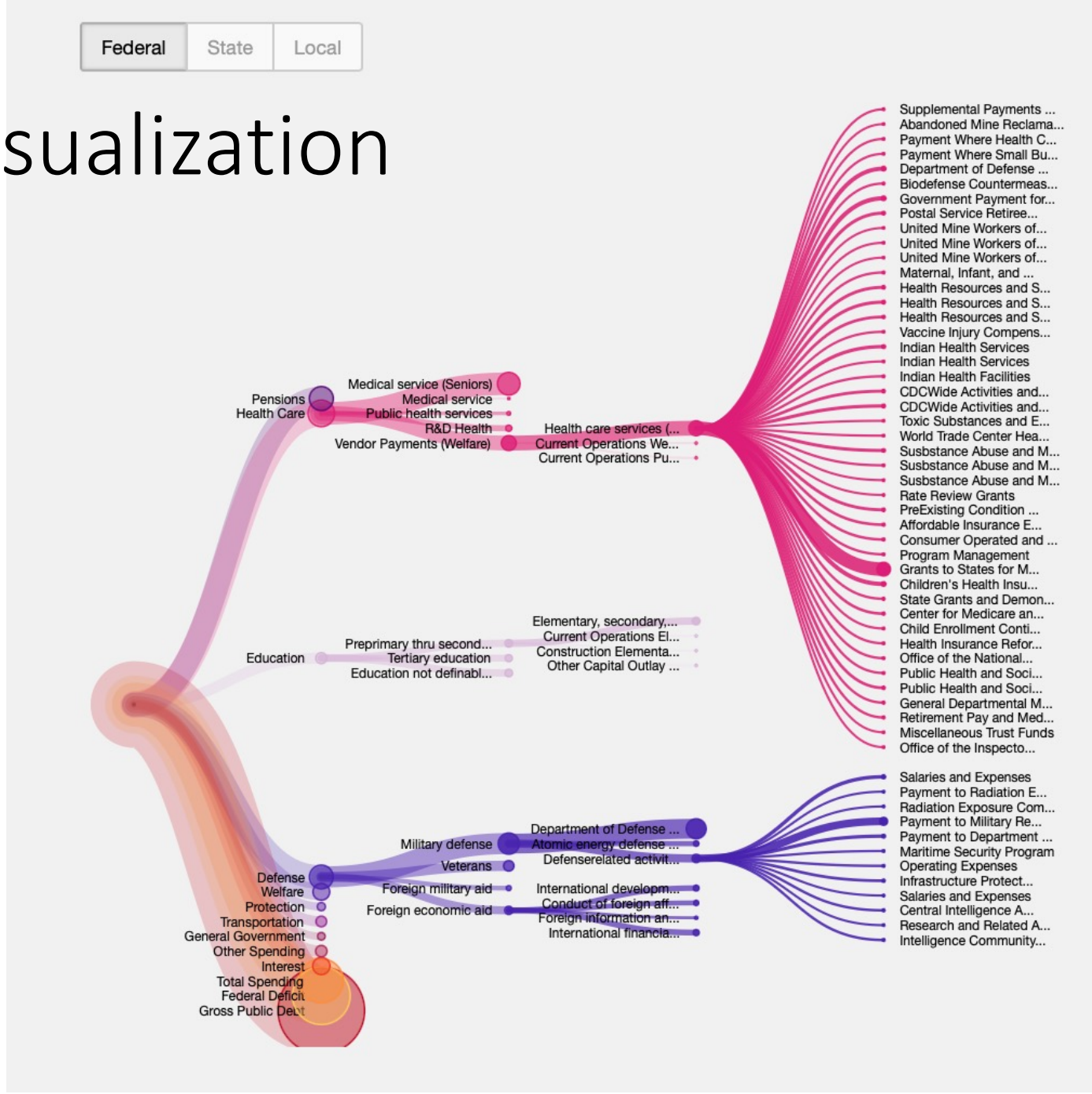


Multi-dimensional Visualization

- Tree/Hierarchical Chart

The goal of this visualization is to allow a user to quickly see the major flows of spending in our government, at the federal, state, and local level. It's interesting to see the relationship of spending between education and defense at both the Federal and Local level.

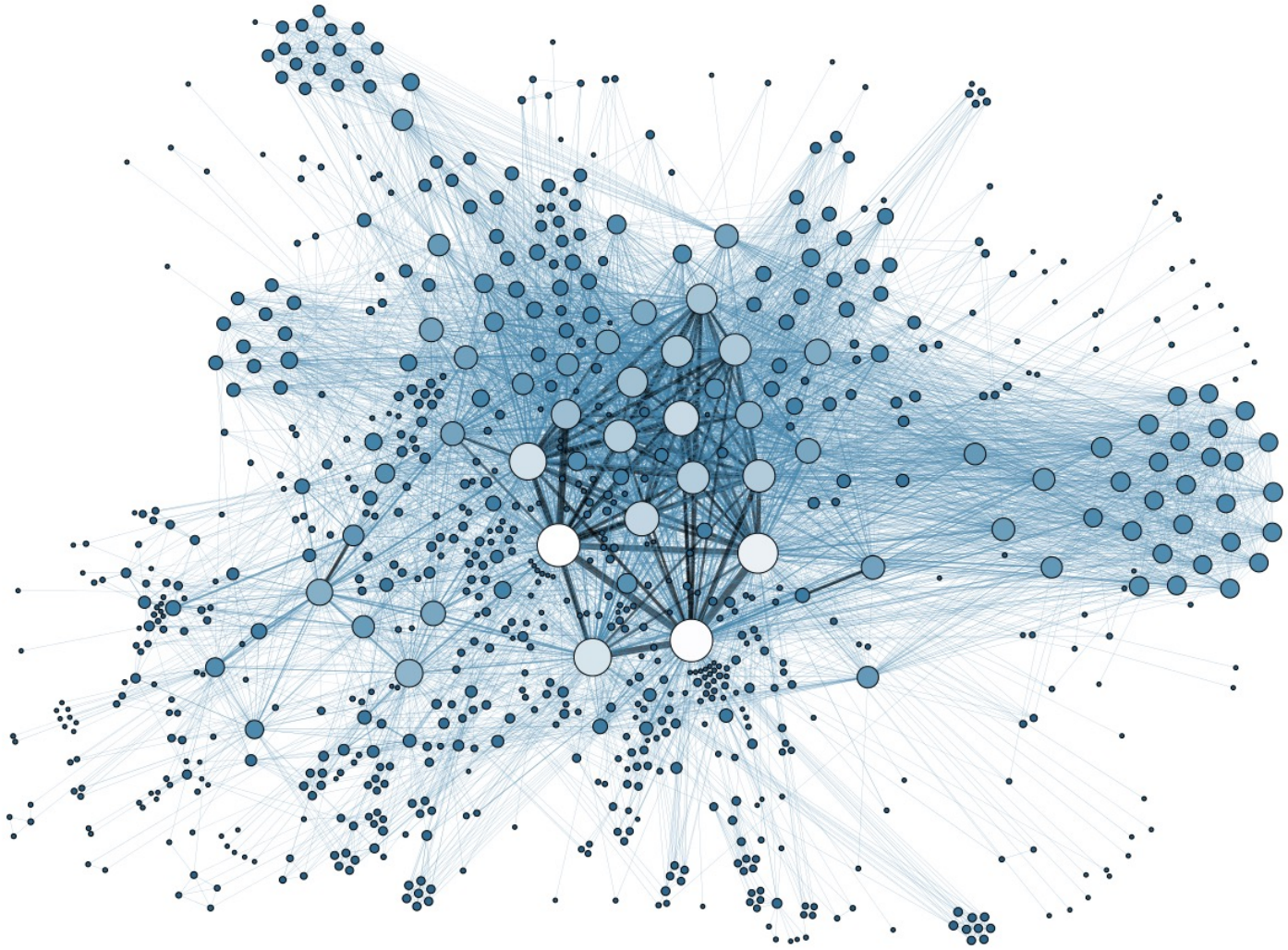
http://www.brightpointinc.com/2013-federal-budget/?source=d3js



Multi-dimensional Visualization

- Network – Social Networks

Graph representing the metadata of thousands of archive documents, documenting the social network of hundreds of League of Nations personals. Grandjean, Martin (2014). "La connaissance est un réseau". *Les Cahiers du Numérique* **10**(3): 37-54. DOI:10.3166/LCN.10.3.37-54.

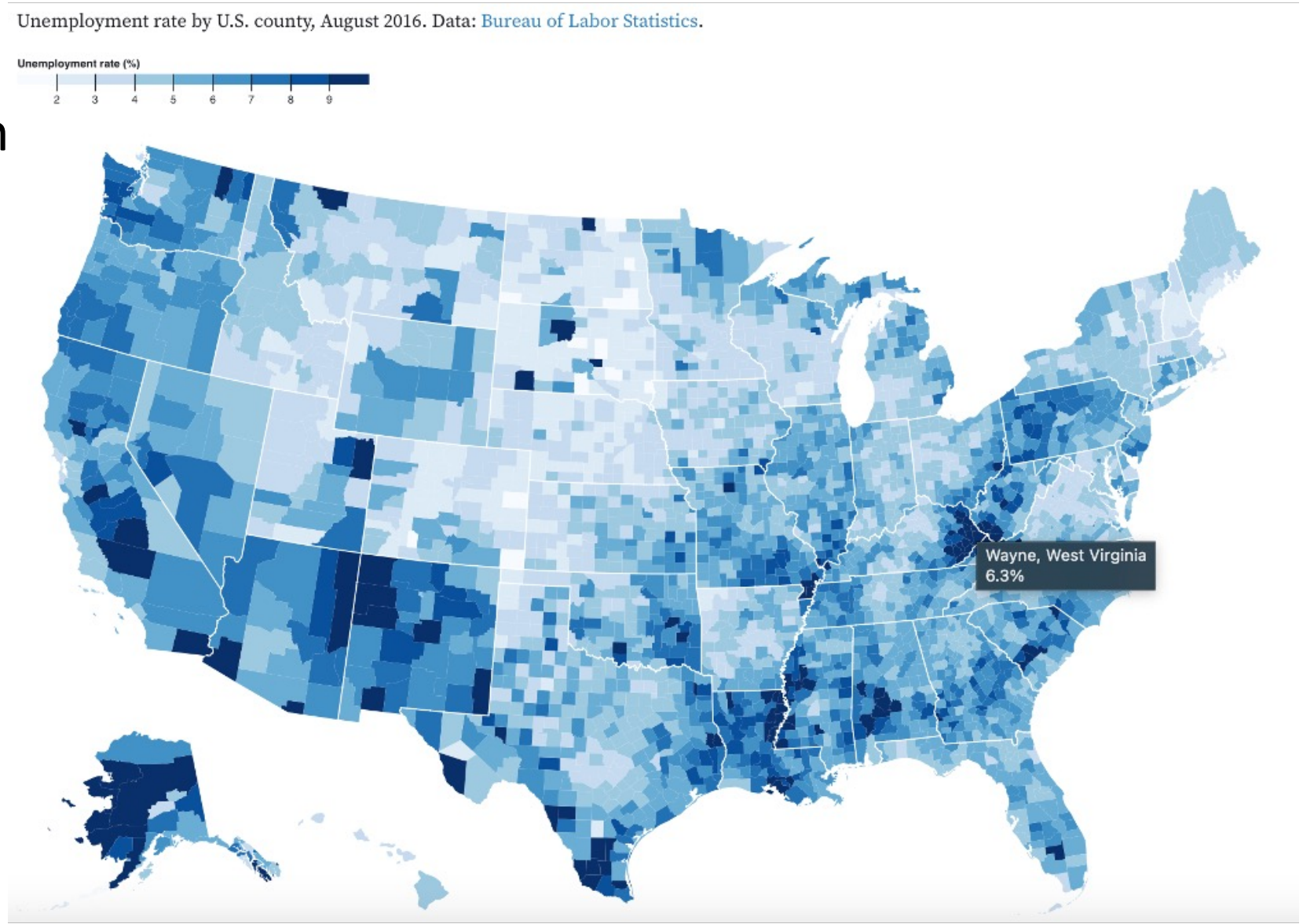


https://commons.wikimedia.org/wiki/File:Social_Network_Analysis_Visualization.png#/media/File:Social_Network_Analysis_Visualization.png

http://mbostock.github.io/d3/talk/20111116/force-collapsible.html

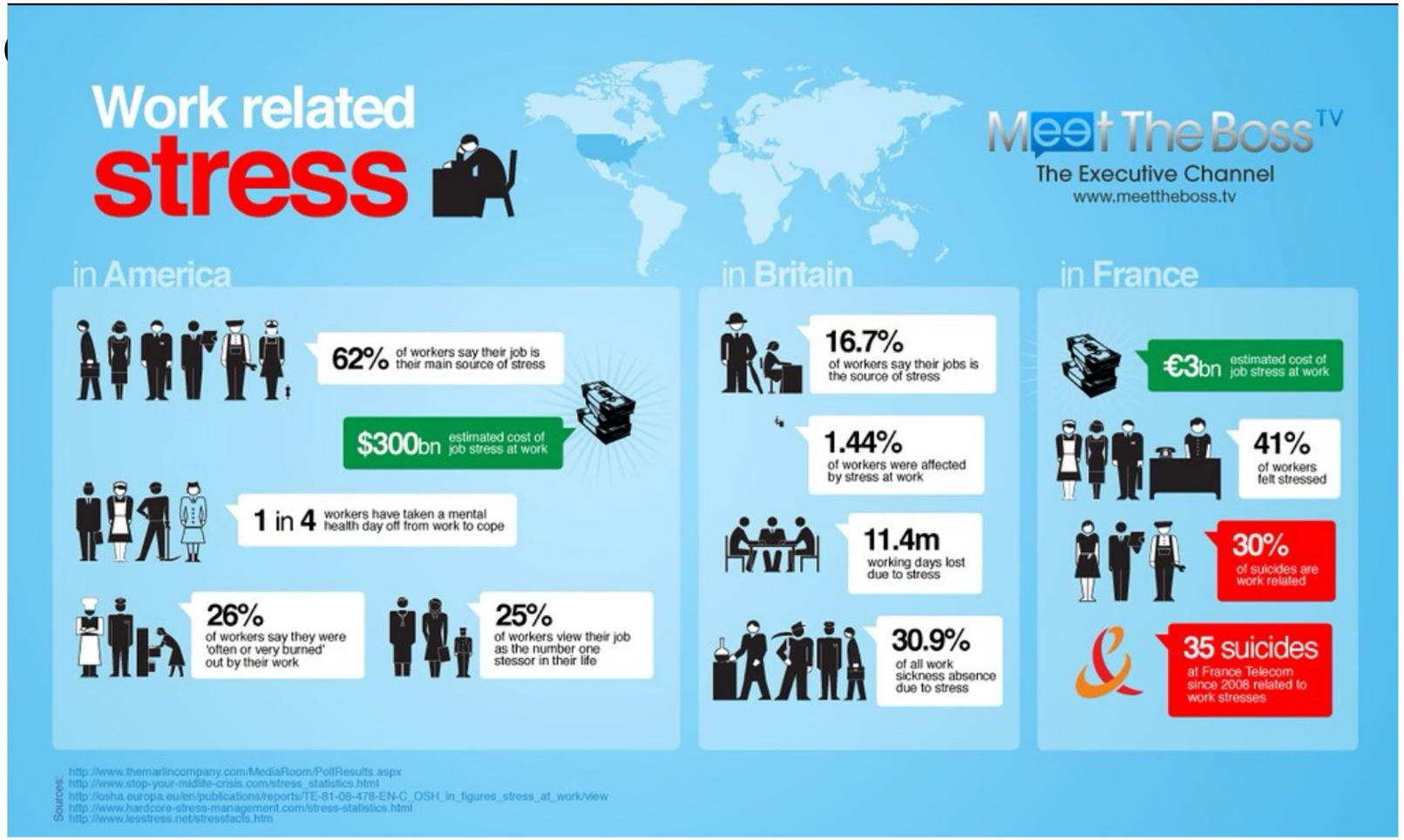
Multi-dimensional Visualization

- Choropleth



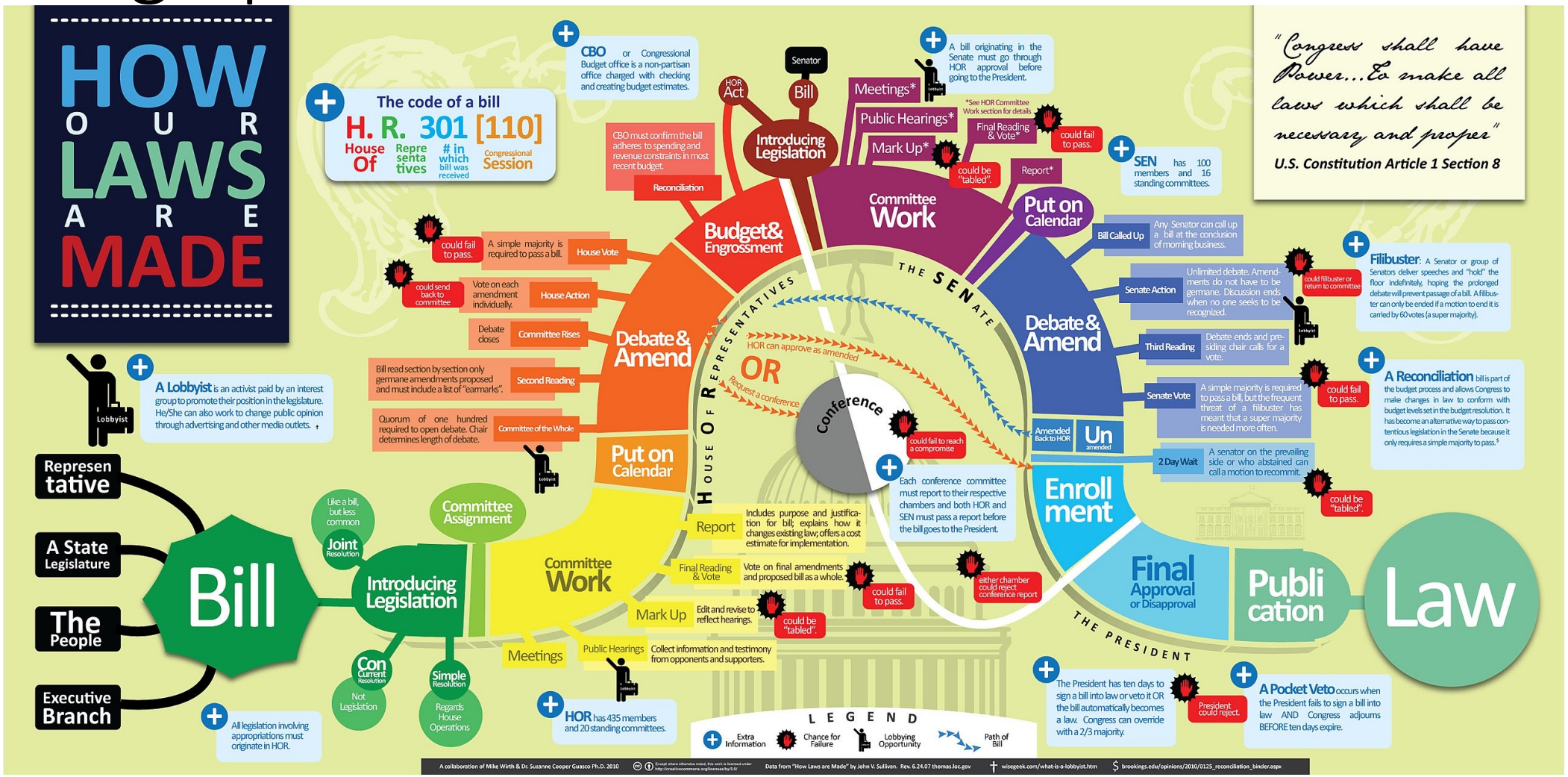
Multi-dimensional Visualization

- Infographi



https://www.flickr.com/photos/gdsdigital/4514262548

Inforgraphic



This “How Our Laws Are Made” infographic by Mike Wirth and Dr. Suzanne Cooper-Guasco won first prize in the Sunlight Foundation “Design for America Competition” 2010. Accuracy was judged based on the details in this document: http://thomas.loc.gov/home/lawsmade.toc.html . Sunlight Foundation “Design for America Competition” 2010: http://sunlightlabs.com/contests/designforamerica/ http://sunlightlabs.com/blog/2010/design-america-winners/ This submission was licensed as Creative Commons By-Attribution 3.0. http://www.ritholtz.com/blog/wp-content/uploads/2010/07/howlawsmadeWIRTH2.jpg http://www.mikewirthart.com/?projects=how-our-laws-are-made
https://commons.wikimedia.org/wiki/File:Visualization-of-How-a-Bill-Becomes-a-Law_Mike-WIRTH.jpg#/media/File:Visualization-of-How-a-Bill-Becomes-a-Law_Mike-WIRTH.jpg

Multi-dimensional Visualization

- Wordcloud
  https://www.jasondavies.com/wordcloud/#//www.jasondavies.com/wordtree/cat-in-the-hat.txt
- Temperature Anomalies
  http://philogb.github.io/page/temperature-anomalies/
- Matplotlib in Python
  https://matplotlib.org/stable/gallery/index.html

Python Tools

- Matplotlib – We all know and love this!
- Mayavi – 3D scientific data visualization
- NetworkX – Graph processing and visualization
- Pyglet – Cross domain multimedia library
- Vispy – High Performance OpenGL based interactive visualization library
- Bokeh – Python interactive visualization in the style of D3.js
- plot.ly – Online plotting and data processing

Note